Universitat de Barcelona

Master in Quantum Science and Technology

Master's thesis

Universal properties from quantum many-body dynamics

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The Loschmidt echo—the amplitude for a quantum state to return to itself—provides a route to the universal properties of a critical many-body system. Continued from real to imaginary time, it becomes the partition function of a conformal field theory on a strip, and its universal data—the central charge and the operator content—are encoded in the spectrum of a transfer matrix, obtained by contracting the space–time tensor network along the space direction. We apply this framework to a transverse-field Ising model extended by a next-nearest-neighbour coupling: a non-integrable chain that remains critical and in the Ising universality class over a wide range of that coupling. After a quench to the critical point, the temporal entanglement that governs the cost of the transverse contraction grows only logarithmically with the evolution time, so the method remains efficient at times where conventional evolution algorithms are limited by the entanglement barrier. We develop a translation-invariant matrix product operator, accurate to second order in the time step, representing the time evolution of the model, together with a block power method that resolves the leading spectrum of the resulting non-Hermitian transfer matrix. Where the method converges, we recover the central charge and operator content predicted by the Ising universality class.

Keywords: Loschmidt Echo, Quantum Quench, Tensor Networks, Transverse Contraction, Temporal Entanglement, Conformal Field Theory, Criticality

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Note on the use of Artificial Intelligence (AI)

Generative AI tools were used during the preparation of this thesis. Anthropic's Claude models assisted with brainstorming, preliminary drafts, debugging numerical code, and automating repetitive tasks, including the production of figures and diagrams. OpenAI's ChatGPT and Google's Gemini assisted with the organisation and clarity of the text and with identifying typographical errors. All AI-assisted material was reviewed, verified, and revised by the author, who takes full responsibility for the content of the thesis.

Contents

Acronyms

CFT	Conformal Field Theory	1
DMRG	Density Matrix Renormalization Group	1
FSM	finite-state machine	III
JW	Jordan–Wigner	9
MPO	Matrix Product Operator	2
MPS	Matrix Product State	1
NN	nearest-neighbour	11
NNN	next-nearest-neighbour	2
RDM	Reduced Density Matrix	7
RTM	Reduced Transition Matrix	III
SVD	Singular Value Decomposition	28
TN	Tensor Network	1
tMPO	temporal Matrix Product Operator	6
tMPS	temporal Matrix Product State	13

1 Introduction and Motivation

Understanding the evolution of a closed quantum system driven far from equilibrium is a central problem in modern many-body physics. Although such dynamics are difficult to simulate, they can reveal universal information at a critical point. In particular, the real-time evolution of a simple initial state can encode the central charge and operator content of the underlying conformal field theory [1], even though these are equilibrium properties of the low-energy spectrum, normally extracted from the ground state. This route is especially valuable for non-integrable models, where analytical results are scarce.

Exact numerical simulations are severely limited by the exponential scaling of the Hilbert space, which makes representing a generic many-body wavefunction intractable. To circumvent this bottleneck, the community has turned to Tensor Networks (TNs), a framework that efficiently compresses quantum states by using entanglement as the organising principle. For one-dimensional systems, Matrix Product States (MPSs) combined with algorithms such as the Density Matrix Renormalization Group (DMRG) have become the state-of-the-art approach for computing ground states [2–4]. Their efficiency comes from the area law of entanglement, which holds for ground states of one-dimensional systems with local gapped Hamiltonians: the entanglement entropy across a bipartition saturates to a constant rather than growing with the subsystem size [5]. Since the required MPS bond dimension is controlled by this entanglement, the ansatz naturally targets the low-entanglement corner of Hilbert space where these ground states lie.

Simulating dynamics, however, runs into the so-called entanglement barrier [6, 7]. After a quantum quench, the entanglement entropy typically grows linearly in time, approaching volume-law behaviour [8]. Although highly optimised algorithms such as time-evolving block decimation (TEBD) [9] and the time-dependent variational principle (TDVP) [10] are effective at short times, capturing this growth requires a bond dimension that increases exponentially with time, making conventional simulations intractable at late times [11, 12].

Transverse contraction [13–15] tackles this problem by adopting a different perspective: rather than evolving a state forward in time as in the Schrödinger picture, it represents the full dynamics as a two-dimensional tensor network and contracts it along the original spatial direction. The computational cost is then controlled by the temporal entanglement entropy [16–19], defined through a Schmidt decomposition across a cut in time rather than in space. Although its physical interpretation remains an open question [20], it is computationally well defined. For a generic quench, temporal entanglement may grow as rapidly as spatial entanglement, but transverse contraction can still be useful for specific dynamical quantities.

The central quantity we compute throughout is the Loschmidt echo for an infinite, translation-invariant system: the amplitude for a state to return to itself after evolving for a time T [21]. It measures how quickly the evolved state departs from, and how coherently it returns towards, its initial condition, providing a natural probe of relaxation after a quench. Because the echo is a single space–time amplitude, transverse contraction computes it efficiently, contracting the network directly with a power method [1].

At a conformally invariant critical point, the echo also provides direct access to universal data such as the central charge of the underlying Conformal Field Theory (CFT). In the transverse picture, it is governed by a spatial transfer matrix whose spectrum is predicted by the CFT and which becomes effectively unitary at long times—a phenomenon known as emergent dual unitarity [1, 22]. Both of these predictions have been verified for the critical Ising and three-state Potts chains [1], along with the corrections that arise because a simulation reaches only finite times [23].

This thesis extends that analysis to the non-integrable regime, studying a self-dual quantum spin chain of the axial next-nearest-neighbour Ising (ANNNI) type in the thermodynamic limit: the transverse-field Ising model with an added coupling of strength p between second neighbours [24, 25]. For any $p > 0$ the chain is longer-ranged and no longer solvable, yet it remains critical and in the Ising universality class over a wide range of p, so the CFT predictions provide a clear target for numerical simulations. We prepare the system in the fully polarised paramagnet state, quench it to the critical point, and use the transverse contraction machinery to compute the Loschmidt echo and the generalised temporal entropies, which we compare with the CFT predictions. This requires two numerical developments. First, we construct a translation-invariant time-evolution Matrix Product Operator (MPO) that is second-order accurate and treats the next-nearest-neighbour (NNN) interactions without requiring prohibitively small time steps. Second, we develop a block power method to resolve the leading transfer-matrix eigenvalues, from which the boundary operator content is obtained.

The thesis is organised as follows. Section 2 introduces the Loschmidt echo and the CFT predictions for the transfer-matrix spectrum and temporal entropies. Section 3 presents the transverse-contraction framework, Section 4 the ANNNI-type model and its equilibrium properties, Section 5 the time-evolution MPO, and Section 6 the two iterations used to extract the transfer-matrix spectrum. Section 7 compares the numerical results with the CFT predictions. The appendices provide supporting derivations, implementation details, numerical controls, and additional measurements.

2 Universal Data from the Dynamics

At a continuous critical point, a lattice model forgets its microscopic details and is described by a CFT, characterised by a small set of universal numbers: the central charge c and the scaling dimensions of its operators. Every model in the same universality class reproduces them, which is what makes them good targets for a numerical method. This section relates the Loschmidt echo to a boundary CFT, derives the four predictions that Section 7 tests, and separates them from the non-universal lattice quantities, which we remove or measure independently.

2.1 From the Loschmidt echo to a boundary CFT

The object we compute is the Loschmidt echo,

$$\mathcal{L}(T) = \langle \Psi_0 | \, e^{-iHT} \, | \Psi_0 \rangle \, . \tag{1}$$

For an infinite, translation-invariant chain prepared in $|\Psi_0\rangle$, this is the return amplitude after evolving for a time T with the critical Hamiltonian. We take $|\Psi_0\rangle$ to be a product state, whose choice fixes the boundary condition discussed below.

To connect this amplitude with the continuum theory, we rotate the evolution time to the imaginary axis, $T \to -i\tau$ (a Wick rotation [26]). The real-time propagator e^{-iHT} then becomes the Euclidean weight $e^{-\tau H}$, the same operator that generates the thermal partition function of a two-dimensional statistical system. The matrix element in Eq. (1) can consequently be written as a Euclidean path integral on a strip of width τ, with both edges fixed by $|\Psi_0\rangle$. A state that can terminate a path integral in this way, without introducing any scale of its own, is what a CFT calls a conformal boundary state [27]. The echo is therefore the partition function of the critical theory on a strip whose width is the (imaginary) evolution time and whose edges carry a conformal boundary condition.

The rotation is used only to obtain the conformal predictions, which are derived on the Euclidean strip; they are continued back to real time before being compared with the simulation. Figure 1 shows this construction, first as a tensor network and then as the strip it represents.

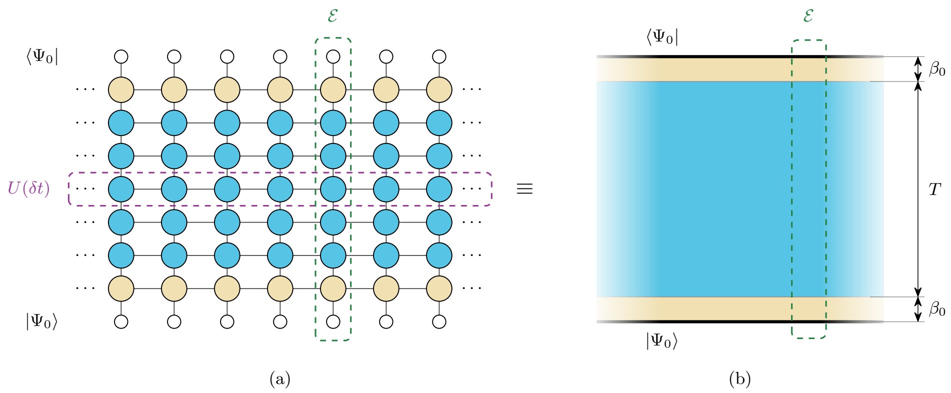


Figure 1: **The Loschmidt echo as a tensor network and as a conformal strip.** (a) Trotterising the evolution with a step δt expresses the echo as the contraction of a two-dimensional network. Open circles are the product state closing the network at the two ends of the time direction; each row of filled circles is one Trotter layer $U(\delta t)$, and the cream rows at top and bottom represent the imaginary-time cooling β_0 that brings the bare product state towards a conformal boundary condition. (b) The same object after the Wick rotation, in the continuum. The heavy lines are the edges where $|\Psi_0\rangle$ closes the path integral, the cream bands are the cooling intervals and the blue bulk is the real-time evolution, so the strip spans $T + 2\beta_0$ in the time direction and is infinite in the spatial one. In both panels the dashed column marks the transfer matrix $\mathcal{E}$, whose construction is developed in Section 3.

Two features of this construction are important for the numerics. First, a bare product state carries weight on high-energy configurations and is not itself a conformal boundary state. We therefore cool it in imaginary time for a short interval β_0, which suppresses that content and brings the state into the regime where the CFT applies. On the lattice, the cooling contributes $N_\beta = 2\beta_0/\delta t$ imaginary-time sites split between the two ends, so the strip spans $T + 2\beta_0$ in the time direction. Second, the initial state is the boundary condition of the strip: each $|\Psi_0\rangle$ realises a different conformal boundary condition, and that choice selects which operators appear in the transfer-matrix spectrum.

In the Euclidean description, space and imaginary time are coordinates of the same two-dimensional geometry, so either can be chosen as the direction of propagation. In the transverse picture, we exchange their roles and contract the network along the original spatial direction. This defines a transfer matrix $\mathcal{E}$ that propagates from one spatial column to the next while acting on states supported on a temporal slice. Its tensor-network construction is developed in Section 3.

In two dimensions, conformal symmetry constrains both the spectrum of $\mathcal{E}$ and the entanglement across a cut in the strip [28, 29]. These two observables provide independent routes to the universal data; Appendix A summarises the conformal mappings used to derive them.

2.2 The universal predictions

Confining a CFT to an infinite strip of transverse size β shifts its ground-state energy by an amount set by the central charge [30]. The excited levels above this shift are organised by the operator scaling dimensions x_i [31]. The transfer matrix inherits this structure [1],

$$\mathcal{E} = e^{-a\hat{H}} = \exp\Big[-\big(-\kappa + \pi L_0\big)\,g\Big], \qquad \kappa = \frac{\pi c}{24}, \tag{2}$$

where L_0 is the operator whose eigenvalues list the scaling dimensions, κ is the Casimir number, and $g = a/\beta$ is the ratio of the column spacing to the strip width, with $a = 1$ in lattice units. The width is complex, $\beta = v(2\beta_0 + iT)$, because the regulator contributes imaginary time while the quench contributes real time. Substituting this width and expanding for $\beta_0 \ll T$ shows that the universal contributions enter the eigenvalue phases at order $1/T$ but affect their moduli only at order β_0/T^2 (Appendix A.2). A single diagonalisation can therefore determine both c and the dimensions x_i from the phases.

The second route uses entanglement. In a CFT, the Rényi entropies follow from correlators of twist operators whose scaling dimension is proportional to c [32]. In the transverse setting, the cut separates earlier from later times rather than dividing space, and its two sides do not belong to a single quantum state. The entropies are instead built from the left and right temporal MPSs of Section 3; they are therefore generalised entropies [20] and are complex in general. For a cut at time t [1, 23],

$$S_n^{\text{gen}}(t) = s_0 + \frac{i\pi c}{24}\Big(1 + \frac{1}{n}\Big) + \frac{c}{12}\Big(1 + \frac{1}{n}\Big)\,W(t, T), \tag{3}$$

with the chord variable

$$W(x, \ell) \;=\; \log\left[\frac{2\ell}{\pi}\,\sin\frac{\pi x}{\ell}\right], \tag{4}$$

where x is the position of the cut and ℓ is the extent of the system. Evaluating W on the interval $(0, T)$ makes the evolution time play the role of the subsystem size in the spatial formula. Because the quench is critical, the entropy grows only as $\log T$, and the bond dimension required for the contraction grows only polynomially with T [11, 12]. We use the Rényi-2 member throughout because it requires only two copies of the system, which also makes it experimentally accessible [33]. The equilibrium calculations of Section 4 instead use the ground-state von Neumann entropy, which DMRG provides directly from the entanglement spectrum.

Equations (2) and (3) now yield four predictions.

Emergent dual unitarity. Equation (2) fixes how the spectrum evolves with T, and predicts that the moduli of the leading eigenvalues approach a common value while their phases stay distinct (see Appendix A.2 for details),

$$\frac{|\mu_i(T)|}{|\mu_0(T)|} = 1 - \frac{b_i}{T^2} - \cdots \xrightarrow[T\to\infty]{} 1, \qquad b_i = \frac{2\pi\beta_0\,(x_i - x_0)}{v}. \tag{5}$$

This indicates that the temporal transfer matrix becomes unitary up to an overall non-universal normalisation [1, 22]. We show this behaviour through the trajectory of $\mu_0(T)$ in the complex plane, which should trace a circle whose coupling-dependent radius reflects the non-universal normalisation.

The central charge from the leading eigenvalue. We work with the logarithms of the transfer-matrix eigenvalues, $\lambda_i = \log(-\mu_i)$, so that $\operatorname{Im}\lambda_i = \arg(-\mu_i)$ is the phase of μ_i measured from the negative real axis. The minus sign shifts every phase by the same constant π; this shift cancels in phase differences and is absorbed by a constant in the fits. The central-charge and boundary-gap relations are expansions in $1/T$ of these logarithms, rather than of the eigenvalues themselves. Substituting the strip width into Eq. (2) and expanding gives the following relation [1], derived in Appendix A.2:

$$\frac{\operatorname{Im}\lambda_0}{T} = A + \frac{B}{T} + \frac{C}{T^2} + \frac{D}{T^4} + \cdots\,, \qquad C = -\frac{\kappa}{v} = -\frac{\pi c}{24\,v}. \tag{6}$$

Here A and D are non-universal coefficients and $B = -\pi$ is the branch constant introduced by $\lambda_0 = \log(-\mu_0)$, while the coefficient of $1/T^2$ carries the Casimir number divided by the velocity. Once v has been measured independently (Section 4.2), the relation $c = 24\, v\, |C|/\pi$ determines the central charge from the eigenvalue phase alone. Because C sits two orders below the extensive term, isolating it needs long time ladders. In practice we fit the unwrapped phase with B pinned to $-\pi$, leaving A and C free.

The boundary dimensions from the phase gaps. Each subleading eigenvalue sits at a phase distance from the leading one that measures a boundary dimension,

$$\mathrm{Im}(\lambda_i - \lambda_0) = \frac{\pi\,(x_i - x_0)}{v\,T}. \tag{7}$$

Once the sound velocity is known, the gaps determine the dimensions; the first of them gives the lightest, x_1. We refer to this spectrum—the dominant eigenvalue followed by one subleading eigenvalue for each boundary operator, ordered by increasing phase distance—as the conformal tower, and to its members as tower states.

Not every dimension appears. Which ones do is fixed by the pair of boundary conditions at the two ends of the strip [34], and for the Ising class there are three conditions: the order parameter may be free to fluctuate, or held fixed in either direction, or a mixed of both. Each pair admits its own set of operators, and each operator brings its own descendants, spaced by integers [27, 35]. The quench of this thesis uses the same free state at both ends, for which $x_0 = 0$ and the spectrum is $\{0, \frac{1}{2}, \frac{3}{2}, 2, \dots\}$. The lightest dimension also controls how quickly the finite-time corrections decay, so the free boundary at $x_1 = \frac{1}{2}$ converges faster than the fixed one at $x_1 = 2$. Appendix G.1 sets out the selection rule for all four pairs and measures each of them.

The generalised entropy profile. Separating Eq. (3) into real and imaginary parts gives

$$\mathrm{Re}\, S_n^{\mathrm{gen}}(t) = s_0 + \frac{c}{12}\Big(1 + \frac{1}{n}\Big)\, W(t,T), \qquad \mathrm{Im}\, S_n^{\mathrm{gen}} = \frac{\pi c}{24}\Big(1 + \frac{1}{n}\Big). \tag{8}$$

The real part follows a symmetric chord profile: plotted against $W(t,T)$ it becomes a straight line whose slope is $\frac{c}{12}(1 + 1/n)$, derived in Appendix A.1. The imaginary part is independent of t, so it delivers c from a single plateau value with no profile to fit. Both parts contain c, but only the plateau gives a usable estimator on the times we can reach (Appendix G.3). Its finite-time correction decays slowly, as $T^{-1/2}$ for the Rényi-2 entropy used here [23], so the plateau must be extrapolated to long times.

These predictions apply to the continuum theory, whereas a simulation contains additional lattice scales and numerical parameters. The sound velocity v, the entropy offset s_0 and the overall normalisation of the transfer matrix are fixed by the lattice construction rather than by the universality class. We measure v independently in Section 4.2, absorb s_0 into the entropy fits, and remove the normalisation by working with phase differences, so that only universal quantities are compared in Section 7. Section 3 constructs the transfer and transition matrices used to make these measurements.

3 Time Evolution via Transverse Contraction

Section 2 showed that the Loschmidt echo of a critical quench encodes conformal data in the spectrum of a transfer matrix acting along the spatial direction of the corresponding strip. We now construct this transfer matrix and explain how it is used numerically. Transverse contraction was introduced for infinite chains by representing the dynamics with MPS extending along the time direction [13, 14], and was subsequently developed to study temporal entropies [1, 18]. Our calculations build on the `ITransverse.jl` implementation [36].

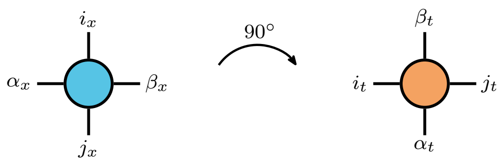
Reading the network along the spatial direction maps the contraction to an eigenproblem for the spatial transfer matrix, whose dominant eigenvectors define the temporal boundary states. These states determine the RTM used to compute the generalised temporal entropies. We finally compare two truncation schemes for controlling the bond dimension of the temporal states.

3.1 The rotated network

The time evolution of a D-dimensional system is represented by a $(D+1)$-dimensional tensor network, with the additional dimension corresponding to time. For the one-dimensional chain studied here, the result is a two-dimensional network with no open physical legs. The echo is a scalar amplitude, so this network is analogous to the partition function of a two-dimensional classical model [37]. Its exact contraction is #P-hard in general [38], so we must use an approximate representation with a controlled bond dimension (Section 3.4).

The tensor-network representation of the echo follows directly from Eq. (1) and is pictured in Figure 1a. The initial and final states are represented as MPSs, while the evolution operator is decomposed into a sequence of MPO layers, one for each Trotter step. Because we require the Hamiltonian to be translation invariant and time independent, the same bulk tensor W appears at every spatial site and time step. Section 5 constructs this tensor for a model with NNN interactions.

In conventional evolution algorithms, successive MPO layers of $U(\delta t)$ are applied to $|\Psi_0\rangle$. The spatial entanglement then grows linearly in time [8], so the bond dimension needed to represent the state grows exponentially [11, 12]. Transverse contraction instead contracts the same network column by column along the spatial direction. The tensors remain unchanged, but their indices exchange roles: the virtual spatial bonds become physical temporal indices, while the physical spatial indices become virtual temporal bonds. Figure 2 illustrates this exchange. The bond dimension of $U(\delta t)$ therefore sets the physical dimension of the temporal problem. Each bulk column defines a temporal Matrix Product Operator (tMPO) $\mathcal{E}$ that acts as a transfer matrix along the original spatial direction, and translation invariance ensures that the same $\mathcal{E}$ is applied at every site.



(a) (b)

Figure 2: **The rotation exchanges physical and virtual indices.** The bulk tensor W read (a) along space, with (i_x, j_x) physical and (α_x, β_x) virtual, and (b) along time, where (α_t, β_t) are physical and (i_t, j_t) virtual.

These objects provide the lattice representation of the strip geometry introduced in Section 2. The strip is infinite along the spatial direction, and applying infinitely many columns on either side of a cut projects the temporal boundary states $\langle b_L|$ and $|b_R\rangle$ onto the dominant left and right eigenvectors of $\mathcal{E}$ [16, 39], which we write $\langle L|$ and $|R\rangle$. These fixed points are independent of how the chain is terminated far from the cut. They encode the temporal boundary condition imposed by $|\Psi_0\rangle$ at the top and bottom edges. The strip width is set by the number of temporal sites, including the real-time evolution and the cooling sites associated with the regulator β_0. The CFT constrains the spectrum of the column transfer matrix $\mathcal{E}$, while the fixed points $\langle L|$ and $|R\rangle$ are the states from which the echo and the temporal entropies are computed.

The row (Trotter-layer) and column transfer operators have different spectral properties. Each row represents the unitary Trotter step $U(\delta t) = e^{-iH\delta t}$ and therefore has eigenvalues of unit modulus, up to the error of its MPO representation. By contrast, the column transfer matrix $\mathcal{E}$ is non-Hermitian and generally has complex eigenvalues with different moduli. The implications of its non-Hermiticity for the left and right eigenvectors are discussed in Section 5 after the operator has been constructed explicitly.

3.2 The dynamics as an eigenvalue problem

Contracting the network at finite spatial size turns the echo into a sum over the spectrum of $\mathcal{E}$. For a chain of N spatial sites,

$$\mathcal{L}(T) \;=\; \langle b_L|\; \mathcal{E}(T)^N \;|b_R\rangle \;=\; \sum_i c_i\, \mu_i(T)^N \;=\; c_0\, \mu_0(T)^N \Big[\, 1 + \sum_{i\geq 1} \frac{c_i}{c_0} \Big(\frac{\mu_i}{\mu_0}\Big)^N \,\Big], \tag{9}$$

where μ_i are the eigenvalues of $\mathcal{E}$, assumed diagonalisable, with left and right eigenvectors $\langle \ell_i|$ and $|r_i\rangle$ normalised so that $\langle \ell_i|r_j\rangle = \delta_{ij}$, and $c_i = \langle b_L|r_i\rangle \langle \ell_i|b_R\rangle$. The dominant pair is $\langle \ell_0| = \langle L|$ and $|r_0\rangle = |R\rangle$. If μ_0 has the largest modulus, every other contribution is suppressed relative to it by $(|\mu_i|/|\mu_0|)^N$. The thermodynamic limit is therefore determined by the leading eigenvalue, and the intensive Loschmidt rate becomes

$$\ell(T) = -\lim_{N\to\infty} \frac{1}{N} \log |\mathcal{L}(T)| = -\log |\mu_0(T)|, \tag{10}$$

up to $\mathcal{O}(1/N)$ boundary corrections for a finite open chain. The calculation thus reduces to a spectral problem whose cost is independent of the spatial length.

Equation (9) reduces the contraction to the spectrum of $\mathcal{E}$. The modulus of μ_0 gives the Loschmidt rate through Eq. (10), while Section 2 reads the central charge from its phase and the boundary dimensions from the phase gaps. The generalised temporal entropies instead require the dominant left and right eigenvectors. Our calculations must therefore resolve the leading eigenpairs of the non-Hermitian operator $\mathcal{E}$, which we obtain with the procedure described in Section 6.

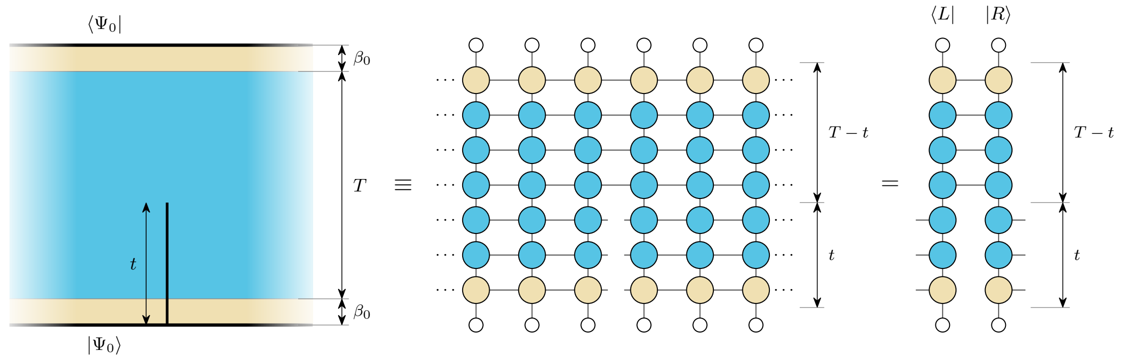
3.3 Temporal states and the RTM

Each application of $\mathcal{E}$ increases the bond dimension of the temporal boundary states. The computational cost is therefore controlled by their temporal entanglement, defined across a cut in the time direction. Section 2.2 showed that after a critical quench the generalised temporal entropies grow logarithmically with the evolution time. We now define the object from which these entropies are computed.

The dominant left and right fixed points of the non-Hermitian transfer matrix are distinct, and the echo depends on their overlap $\langle L|R\rangle$. A Reduced Density Matrix (RDM) formed from either state alone therefore does not contain the required information. The relevant object is instead the RTM constructed jointly from the two temporal boundaries [1, 18],

$$\mathcal{T}_t = \frac{\mathrm{Tr}_{>t}\, |R\rangle\, \langle L|}{\langle L|R\rangle}, \tag{11}$$

where $\mathrm{Tr}_{>t}$ traces over the $T-t$ temporal sites above the cut and leaves the t sites below it open. Thus, $\mathcal{T}_t$ acts on the first t temporal sites. The denominator normalises the matrix so that $\operatorname{Tr}\mathcal{T}_t = 1$. Since $\langle L|$ is not the Hermitian conjugate of $|R\rangle$, $\mathcal{T}_t$ is generally non-Hermitian and may have complex eigenvalues. The consequences of this complex spectrum for truncation are discussed in the next subsection. Figure 3 shows the position of $\mathcal{T}_t$ within the strip geometry.



(a) the strip (b) the tensor network (c) $\mathcal{T}_t$

Figure 3: **Position of the reduced transition matrix.** (a) The cut lies on one spatial bond of the strip and extends from the bottom edge to height t; the cream bands at the two ends are the cooling intervals β_0, and the heavy edges carry the initial state $|\Psi_0\rangle$. (b) In the tensor-network representation, the horizontal bonds across the cut remain open over the first t temporal sites, while the two sides remain contracted over the remaining $T-t$ sites. (c) Contracting the columns to the left and right of the cut gives the temporal boundary states. The power methods of Section 6 converge them to the dominant left and right eigenvectors $\langle L|$ and $|R\rangle$ of $\mathcal{E}$. Their contraction over the upper $T-t$ sites gives $\mathcal{T}_t$ in Eq. (11), with the lower t indices left open.

The spectrum of $\mathcal{T}_t$ defines the generalised temporal entropies associated with the transition between the two boundary states. We use the Rényi-2 member of the family introduced in Section 2,

$$S_2^{\text{gen}}(t) = -\log \operatorname{Tr}\left(\mathcal{T}_t^2\right). \tag{12}$$

When $\langle L|$ and $|R\rangle$ differ, the local reduction of $\mathcal{T}_t$ preserves its singular values but not its eigenvalues, so its spectrum cannot be read from a local decomposition (Appendix B). This is the member we can compute: $\operatorname{Tr}(\mathcal{T}_t^2)$ is evaluated by contracting two copies of the RTM, without diagonalising it. The finite-time corrections to S_2^{gen} must be accounted for when comparing with the conformal predictions (Section 2.2).

Once $\langle L|$ and $|R\rangle$ have converged for a target evolution time T, we evaluate Eq. (12) at every internal bond of the temporal chain, corresponding to cuts $t \in (0, T)$. The entropy is generally complex because $\mathcal{T}_t$ is non-Hermitian, so its real and imaginary parts are analysed separately as functions of the normalised cut position t/T.

The region represented by $\mathcal{T}_t$ touches the lower edge of the strip, so only one of its endpoints lies in the interior. This halves the logarithmic coefficient in Eq. (8) relative to an interval with both endpoints in the bulk (Appendix A.1). The entropy-route central charges reported in Section 7 are extracted using this geometry.

3.4 Truncation based on RDM and RTM

We use two complementary truncation schemes based on the RDM and the RTM. The conventional RDM truncation, available in most tensor-network libraries, brings a boundary MPS into mixed canonical form and discards the smallest singular values of its orthogonality centre, giving the optimal low-rank approximation to that state [3, 40]. The gauge-invariant scheme implemented in `ITransverse.jl` [36] instead truncates the RTM of Eq. (11), which depends on both temporal boundaries.

The difference between both schemes is in what they consider as relevant. The RDM keeps the directions in which $|R\rangle$ or $\langle L|$ has the largest weight, regardless of their contribution to $\langle L|R\rangle$. The RTM instead weights each direction by its contribution to this overlap, so a large component of $|R\rangle$ may have little importance if it is orthogonal to the corresponding component of $\langle L|$ and can be discarded [18, 39]. The RTM therefore reaches a given accuracy at a smaller bond dimension, with the RDM truncation being approximately an order of magnitude more expensive in our calculations (Appendix F.6).

This reduction in cost comes at the expense of not having a variational guarantee for RTM truncation. For a positive Hermitian RDM, the eigenvalues are the squared Schmidt coefficients and coincide with its singular values, so retaining the largest gives the optimal approximation at fixed bond dimension. The RTM on the other hand is non-Hermitian: its eigenvalues may be complex or negative and no longer correspond one to one with its singular values, so the truncation criterion carries no variational bound. We therefore use the RTM scheme for production calculations and retain the RDM scheme as a complementary check; Appendix F.6 verifies that the two agree at every evolution time where both were run. Appendix B presents both truncations in detail, including their behaviour under gauge transformations and the conditions for optimality.

4 The ANNNI-type Model

We study a self-dual quantum spin chain that extends the transverse-field Ising model with NNN interactions [25]. For spin-1/2 degrees of freedom, the Hamiltonian is

$$H = -\sum_i \Big[\sigma_i^z \sigma_{i+1}^z + \lambda\, \sigma_i^x \;+\; p\left(\sigma_i^z \sigma_{i+2}^z + \lambda\, \sigma_i^x \sigma_{i+1}^x\right)\Big], \tag{13}$$

where $\sigma^{x,z}$ are Pauli matrices, λ is the transverse field, and p controls the strength of the additional interactions. The $\sigma^x\sigma^x$ term preserves self-duality under the Kramers–Wannier transformation [25], which fixes the critical point at $\lambda = 1$. We work at this point throughout, so that the conformal predictions of Section 2 apply.

At $p = 0$, Eq. (13) reduces to the integrable transverse-field Ising chain. For $p > 0$, a Jordan–Wigner (JW) transformation no longer maps the Hamiltonian to free fermions, and the model is therefore non-integrable. The origin of the fermionic interactions is detailed in Appendix C.

4.1 Equilibrium criticality and the central charge

We restrict our equilibrium calculations to $0 \le p \le 1.5$, the range over which the model is known to remain critical and in the Ising universality class [25]. We verify the equilibrium universality class over this range using the ground-state entanglement. For a critical open chain of length L, CFT predicts that the von Neumann entropy of a block of length l scales as [32, 40, 41]

$$S_{\mathrm{vN}}(l, L) = \frac{c}{6}\, W(l, L) + k, \tag{14}$$

where W is the chord variable of Eq. (4) and k is a non-universal constant. The von Neumann entropy is well suited to this calculation because DMRG provides the full ground-state Schmidt spectrum directly, and also has weaker finite-size corrections than the higher-order Rényi entropies [42, 43].

For each value of p, we obtain the ground state of a chain with $N = 400$ sites and evaluate the entropy across every bond. Equation (14) predicts a linear dependence on $W(l, L)$, so the entanglement profile of a single ground state provides an estimate of the central charge. We fit only the middle part of the chain, using the cuts satisfying $0.25L < l < 0.75L$, where boundary corrections are smallest. The resulting profiles are shown in Figure 4.

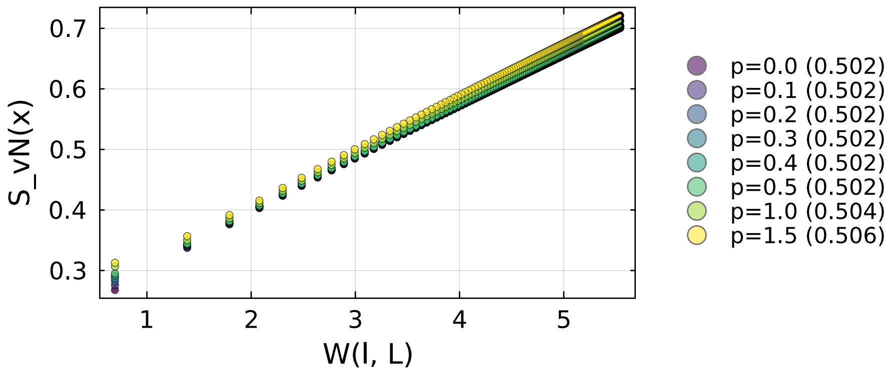


Figure 4: **Effect of NNN interactions on the equilibrium central charge.** Ground-state entanglement profiles at $\lambda = 1$ and $N = 400$, for the eight couplings $p = 0$ to 1.5. Solid lines are fits over the middle half of the chain, and the legend gives the central charge each fit returns. The linear dependence on W gives a value close to the Ising $c = 1/2$ at every p, with a small common offset from finite-size corrections.

The profiles follow the conformal scaling law at every coupling, and the fitted central charge barely changes with p. Its small common offset from the exact value $c = 1/2$ is consistent with residual finite-size corrections to Eq. (14), since a similar offset is already present at the exactly solvable point $p = 0$. The additional interactions therefore do not change the universality class over the range considered. This agrees with previous finite-size scaling studies of the same chain [25], and with an independent determination that extracts the conformal data directly from the lattice Hamiltonian [44].

The profiles separate at small W, where the cuts lie closest to the open boundaries. Equation (14) is less accurate in this region because the boundaries introduce corrections that depend on the microscopic interactions and therefore on p, which is why these cuts are left out of the fit. Towards the centre of the chain, these boundary effects weaken and the profiles converge towards the common linear behaviour predicted by CFT. Appendix D.1 gives an independent estimate from the finite-size scaling of the mid-chain entropy and a cross-check using the Rényi-2 entropy.

4.2 Sound velocity

The NNN interaction changes how fast excitations travel along the chain. At an Ising-class critical point, the low-energy dispersion is linear, $\omega(k) = v|k|$, so the sound velocity v fixes the relativistic scale of the continuum theory [29]. It enters three of the four predictions of Section 2 explicitly, because those predictions are written in terms of lengths on the conformal strip and v is what turns an evolution time into such a length [30, 31]; in the entropy profile it only shifts the non-universal offset. The velocity must therefore be measured independently before the predictions can be compared with the dynamical data.

At $p = 0$ the chain is a free-fermion model with the exact dispersion $\omega(k) = 4|\sin(k/2)|$, hence $v = 2$ [45]; for $p > 0$ we determine v numerically.

We extract $v(p)$ from the equilibrium dispersion of the same chain, Eq. (13) at $\lambda = 1$, placed on rings of N sites with periodic boundary conditions and diagonalised exactly. The momenta on a ring are quantised, so the slope follows from the energy of the lowest excitation carrying the smallest non-zero momentum [25]. Because exact diagonalisation is restricted to small rings, we repeat this at five lengths and extrapolate with a $1/N^2$ dependence, a form fixed empirically at the integrable point where the answer is known (Appendix D.2). This procedure reproduces the exact result at $p = 0$ and is then applied at each non-zero coupling p.

The velocity grows steadily from 2.000 at $p = 0$ to 10.799 at $p = 1.5$, as shown in Figure 5. Although the velocity is non-universal, it is fixed by the model; we determine it from the equilibrium spectrum and use the independently extrapolated value at each coupling in all subsequent conversions. Appendix D.2 describes the dispersion calculation, extrapolation and validation in detail.

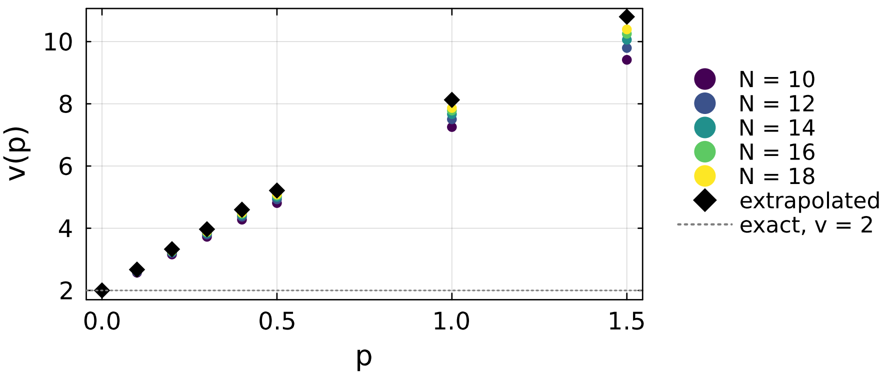


Figure 5: **The sound velocity grows with the NNN coupling.** $v(p)$ at $\lambda = 1$, from the equilibrium dispersion of periodic rings of $N = 10$ to 18 sites (coloured circles), together with the values extrapolated to the thermodynamic limit (black diamonds), which are the ones used throughout. The dotted line marks the exact Ising value $v = 2$ at $p = 0$.

5 Building the Time-Evolution MPO

For nearest-neighbour (NN) Hamiltonians, the evolution operator $U(\delta t) = e^{-iH\delta t}$ is usually represented as a Trotter–Suzuki circuit of two-site gates, giving rise to the usual brick-wall structure [12]. However, an interaction between sites i and $i + 2$ cannot be assigned to a single bond within this pattern. To preserve the one-site translation invariance required by the transverse contraction, as discussed in Section 3.1, we instead express the Hamiltonian as a finite-state machine (FSM) and exponentiate the resulting representation directly.

5.1 The Hamiltonian as a FSM

In the FSM representation [46, 47], the Hamiltonian is encoded as an MPO whose local tensor has a block-upper-triangular form [48]. Two identity blocks propagate the initial and final states, while the on-site block D generates local terms. The block row C initiates interactions, the block column B terminates them, and the memory block A

propagates unfinished interactions between their endpoints. The virtual index therefore records whether an interaction has not yet begun, is being propagated, or has terminated. For more complicated Hamiltonians, this block structure can be generated automatically from the operator [49]. For the ANNNI-type model of Eq. (13), the local tensor has bond dimension $D_W = 5$ and takes the form

$$W_H = \begin{pmatrix} \mathbb{1} & C & D \\ 0 & A & B \\ 0 & 0 & \mathbb{1} \end{pmatrix} = \begin{pmatrix} \mathbb{1} & -\sigma^z & -p\lambda\,\sigma^x & -p\,\sigma^z & -\lambda\,\sigma^x \\ 0 & 0 & 0 & 0 & \sigma^z \\ 0 & 0 & 0 & 0 & \sigma^x \\ 0 & \mathbb{1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathbb{1} \end{pmatrix}. \tag{15}$$

Figure 6 represents Eq. (15) through the allowed paths between the five virtual states. Each node denotes a value of the virtual index, while an arrow from state a to state b, labelled by an operator O, represents the tensor entry $(W_H)_{ab} = O$. The FSM moves from left to right along the chain, beginning in state 1 and ending in state 5.

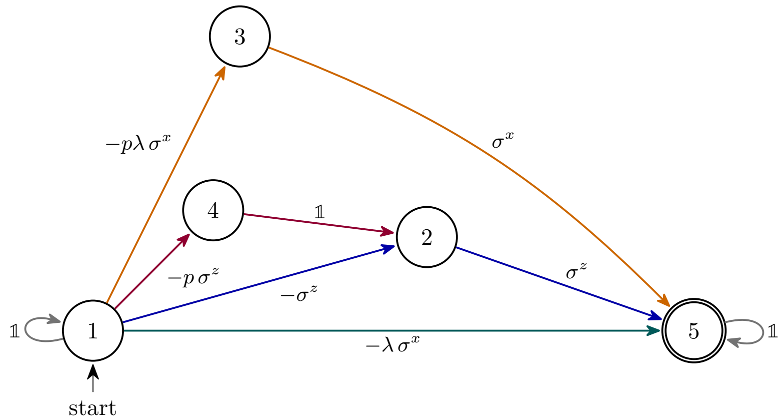


Figure 6: **The Hamiltonian as a FSM.** The nodes are the five virtual states of Eq. (15), and every path from state 1 to state 5 (double circle) generates one term of the Hamiltonian, applying the operator that labels each arrow. Colours mark which term is generated: the transverse field in teal, the NN couplings in blue ($\sigma^z\sigma^z$) and orange ($\sigma^x\sigma^x$), and the NNN coupling in purple. Grey loops apply identities before and after the term.

The loop $1 \to 1$ applies identities until a Hamiltonian term is initiated. A direct transition $1 \to 5$ applies the on-site field and completes the term at the same site. An NN interaction instead follows either $1 \to 2 \to 5$ or $1 \to 3 \to 5$: the first transition applies the operator at site i and records whether a σ^z or σ^x endpoint remains to be applied; the second applies the corresponding operator at site $i+1$ and completes the interaction. The NNN term follows $1 \to 4 \to 2 \to 5$: the first transition applies σ^z at site i, the second inserts an identity at site $i+1$, and the third applies σ^z at site $i+2$. After state 5 is reached, the loop $5 \to 5$ applies identities along the remainder of the chain. Zero entries in W_H correspond to absent arrows, so no other transitions are allowed.

Because the FSM is directional, its local tensor is asymmetric under interchange of the virtual indices. The exponentiated operator inherits this structure, as does the spatial transfer matrix after moving to the transverse picture. The resulting operator $\mathcal{E}$ is generally non-Hermitian and non-normal, $\mathcal{E}\mathcal{E}^\dagger \neq \mathcal{E}^\dagger\mathcal{E}$. Its left and right eigenvectors are therefore not related by Hermitian conjugation, and we compute the two temporal boundary states separately.

5.2 Exponentiating the FSM

We seek an approximation $W(\tau) \approx e^{\tau H}$, with $\tau = -i\delta t$, built from the blocks of W_H. Because our Hamiltonian carries interactions at two different ranges, we use the construction of Van Damme et al. [50], which organises the expansion by the spatial overlap of the local Hamiltonian terms. Contributions with disjoint support factorise and are kept to all orders, so only connected clusters of overlapping terms have to be reproduced explicitly. Appendix E.1 shows how the operator is built from the machine of Figure 6.

We use its second-order member, denoted *VD2*, whose virtual bond dimension is 13 for the model of Eq. (13). As noted in Section 3.1, the virtual bond dimension of the time evolution MPO becomes the local physical dimension of the temporal chain after moving to the transverse picture. Consequently, *VD2* carries a larger local dimension than cheaper constructions like the ones of Zaletel et al. [51], which are only first order in a single application. For a given accuracy, the latter thus demand a smaller time step, while *VD2* can hold the same accuracy at a bigger δt. The larger step shortens the temporal chain at fixed physical time and lowers the number of truncation steps, which is what makes the long-time calculations of Section 7 affordable. Appendix E gives the full construction, the explicit tensor, and a measurement of the convergence order of each operator.

The temporal transfer matrix is assembled from a single tensor of the time evolution MPO, repeated along the chain, so the tensor we extract must be one that a site in the bulk would carry, with every interaction present. We therefore exponentiate the Hamiltonian on a chain long enough for its central site to have all of its partners inside it, five sites for an NNN model, and take the tensor at the centre. Every result in this thesis uses this construction.

6 Extracting the transfer-matrix spectrum

Section 3 reduced the dynamical calculation to the problem of determining the spectrum of the transfer matrix $\mathcal{E}$. We use two iterative methods, chosen according to the spectral information required by each observable and their computational cost.

The right and left eigenvectors of $\mathcal{E}$, written $|r_i\rangle$ and $\langle \ell_i|$ as in Section 3, satisfy

$$\mathcal{E}\,|r_i\rangle = \mu_i\,|r_i\rangle\,, \qquad \langle \ell_i|\,\mathcal{E} = \mu_i\,\langle \ell_i|\,. \tag{16}$$

Since $\mathcal{E}$ is non-normal (Section 5.1), its right and left eigenvectors differ and must be computed separately. They form a biorthogonal set, such that $\langle \ell_i|r_j\rangle \propto \delta_{ij}$. Throughout, the bra notation does not imply complex conjugation: $\langle \ell_i|$ denotes a row vector, and $\langle \ell_i|r_j\rangle$ is the direct contraction of the two temporal networks. We implement the second equation by propagating the left states with $\mathcal{E}^{\mathsf{T}}$.

6.1 The single-vector iteration

The first option we use is the power method implemented in `ITransverse.jl` [36]. Starting from a random complex temporal Matrix Product State (tMPS), we repeatedly apply the transfer matrix and normalise after each step. As shown in Figure 7, every application multiplies the temporal bond dimension by the tMPO bond dimension D, so we use either truncation scheme of Section 3.4 to cap the bond dimension to χ, keeping the cost per iteration fixed. The reason why we use random initial states is because a structured product state may have little overlap with the dominant eigenvector and may therefore converge to a subdominant fixed point.

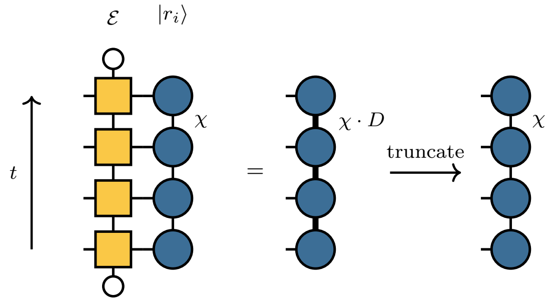


Figure 7: **One application of the transfer matrix to a temporal state.** The tMPS forms a boundary column, while $\mathcal{E}$ is the corresponding column of the space–time network. Applying it raises the temporal bond dimension from χ to χD, where D is the tMPO bond dimension, and the state is then truncated back to χ. The left state is treated in the same way using the transpose.

The iteration converges to the dominant right and left fixed points, from which we obtain the leading eigenvalue as

$$\mu_0 = \frac{\langle L| \mathcal{E} |R\rangle}{\langle L|R\rangle}. \tag{17}$$

Therefore, we use this single-vector method in Section 7 for computing the leading eigenvalue and the associated dominant pair ($\langle L|$ and $|R\rangle$). Obtaining the boundary dimensions x_i, on the other hand, requires access to the subleading eigenvalues of the spectrum to compute the gaps between them.

6.2 The block iteration

To resolve the subleading eigenvalues, we propagate blocks of k right and left states, denoted by $|R_j\rangle$ and $\langle L_j|$. Repeated application of $\mathcal{E}$ drives each block towards the subspace spanned by the k leading eigenvectors, although its individual states do not generally converge to particular eigenvectors and must be separated afterwards. The convergence rate is controlled by the spectral separation between the retained subspace and the discarded eigenvalues, rather than by the gaps within the retained spectrum. The block can therefore converge even as the leading eigenvalues approach the common-modulus structure described in Section 2.2.

The individual eigenvalues are then separated by projecting the eigenproblem onto the converged subspace. We approximate a right eigenvector as a linear combination of the current block states,

$$|\psi\rangle = \sum_{j=1}^{k} v_j \, |R_j\rangle \, . \tag{18}$$

Inserting this approximation into $\mathcal{E}\,|\psi\rangle = \theta\,|\psi\rangle$ produces a residual which we require to vanish when contracted with every member of the left block, $\langle L_i|\,(\mathcal{E}-\theta)\,|\psi\rangle = 0$. This two-sided projection is appropriate for a non-normal operator [52] and reduces the full eigenproblem to two $k \times k$ matrices,

$$S_{ij} = \langle L_i|R_j\rangle \, , \qquad M_{ij} = \langle L_i|\, \mathcal{E} \,|R_j\rangle \, , \tag{19}$$

and to the generalised eigenvalue problem

$$\sum_j M_{ij}\, v_j = \theta \sum_j S_{ij}\, v_j. \tag{20}$$

The overlap matrix S describes the relative orientation of the left and right subspaces, while M is the transfer matrix projected between them. Solving this reduced problem yields approximations θ_j to $\mu_0, \dots, \mu_{k-1}$ and hence the phase gaps that encode the boundary operator content.

The block iteration can also reconstruct the individual eigenvectors, but this step is computationally expensive. Each eigenvector is formed as a linear combination of the propagated block states, which increases the temporal bond dimension and requires a further truncation to χ. This repeated enlargement and truncation dominates the cost of the eigenvector mode. The single-vector method avoids this operation and provides the dominant eigenvectors at a lower cost, so we use it for determining the dominant pair and the block method only for the subleading eigenvalues. Appendix F describes the implementation details of this block power method.

6.3 Selecting and tracking the eigenvalue

The block iteration returns k unlabelled Ritz values. At each time, we must identify μ_0 and assign the remaining values to the conformal tower. Selecting the value of largest modulus is unreliable because, for a non-normal operator, an unconverged subspace can produce spurious Ritz values outside the spectrum, and their number may increase with the block size [52]. We instead identify the physical branch through continuity in time. We retain candidates whose modulus lies within a few per cent of the previously accepted μ_0 and select the nearest of these in the complex plane. If no candidate satisfies this criterion, we discard the time point. The remaining values are ordered by increasing phase distance from μ_0, as predicted for the conformal tower. We compute each ladder at increasing times and initialise every block with the converged result from the preceding time. Appendix F.5 specifies the thresholds and reports the cases in which reselection changed the result.

The single-vector iteration returns one eigenvalue per run, so its reliability is assessed across independent seeds rather than among several Ritz values. At short times, all seeds converge to the same left and right fixed points. At longer times, their overlap $\langle L|R\rangle$ can decrease, making the pair ill-conditioned and allowing different runs to converge to different fixed points. Appendix F.6 examines the onset of this behaviour. We accept an entropy time point only when two independent runs agree and the resulting profile varies smoothly between neighbouring times. To continue a leading-eigenvalue ladder, we apply the continuity criterion across seeds and retain the run whose phase follows the progression established at earlier times. We include the new point only if it does not increase the residual of the fit to Eq. (6). Appendix G.4 describes how the fitted ladders are assembled, while Appendix F.6 reports the failure statistics of both iterations.

7 Results

We test the CFT predictions of Section 2.2 using simulations of the critical ANNNI-type chain, Eq. (13) with $\lambda = 1$, quenched from the polarised state $|X+\rangle$. The NNN coupling ranges from the integrable point, $p = 0$, to $p = 0.5$. Extracting the universal data requires a sufficiently long time window, while the computational cost increases rapidly with p; we therefore restrict the dynamical calculations to this range.

We follow the two routes introduced in Section 6: the entropy route, which requires the dominant eigenvectors, and the spectral route, which uses the leading eigenvalues. Both analyses include the integrable Ising point and the non-integrable cases, allowing us to determine whether the conformal signatures persist after introducing the NNN interaction.

Table 1 summarises the resulting universal data.

Unless stated otherwise, the simulations use the second-order *VD2* MPO of Section 5, with time step $\delta t = 0.1$, cooling regulator $\beta_0 = 0.2$ ($N_\beta = 4$), and maximum temporal bond dimension $\chi = 64$. We convert the spectral measurements into universal quantities using the independently determined sound velocities $v_\infty(p)$ from Section 4.2. Appendix G.4 gives the fitting conventions and numerical controls.

		spectral route			entropy route	
p	v_∞	window	c	x_1	window	c
0	2.000	$T \leq 24$	0.45	0.498	$8 \leq T \leq 24$	0.50
0.1	2.670	$T \leq 17$	0.47	0.497	$4 \leq T \leq 13$	0.50
0.3	3.967	$T \leq 10$	0.49	0.493	$2 \leq T \leq 9$	0.53
0.5	5.212	$T \leq 6$	0.53	0.487	$2 \leq T \leq 5$	0.58

Table 1: **Universal data as a function of the NNN coupling.** The spectral central charge is obtained from the phase curvature of μ_0 through Eq. (6), while x_1 follows from the first phase gap through Eq. (7). The entropy-route central charge is extracted from the extrapolated plateau of Section 7.1, using seed ensembles of independent single-vector runs. Each route has its own fitting window, listed beside its result. Sensitivity to that window dominates the uncertainty of the entropy values and increases as the available ladders shorten. The values at $p = 0.3$ and $p = 0.5$ are therefore indicative rather than precise. Appendix G.4 gives the fitting conventions and controls.

7.1 The entropy route

At the integrable point, the generalised temporal entropy of Eq. (12) follows the conformal prediction previously established for the Loschmidt echo of the critical Ising chain [1]. This behaviour persists after introducing the NNN interaction. Figure 8 shows the Rényi-2 profiles at $p = 0.1$: the real part follows the chord dependence of Eq. (8), while the imaginary part develops a plateau across the interior of the strip. In this comparison, we fix $c = \frac{1}{2}$ and fit only the non-universal constant s_0. Appendix G.2 presents the corresponding profiles at the other couplings.

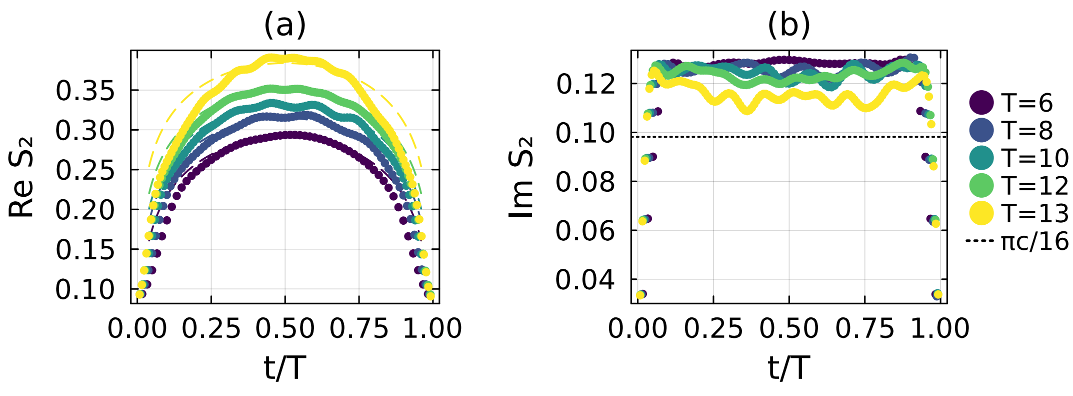


Figure 8: **Generalised temporal entropy at $p = 0.1$.** Each colour denotes one evolution time. (a) Real part as a function of t/T; the dashed curves show the conformal profile of Eq. (8) with $c = \frac{1}{2}$. (b) Imaginary part, which forms an approximately constant plateau in the interior of the strip. The plateau lies above the asymptotic value $\pi c/16$ (dotted line) and approaches it as T increases.

At all simulated times, the measured plateau lies above the asymptotic value $\pi c/16$, consistent with the finite-time correction introduced in Section 2.2 [23]. Determining the asymptote therefore requires an extrapolation over several evolution times, whose reliability depends on the length of the available sequence. Within a short window, the predicted decay and a constant offset describe the data comparably well but extrapolate to different limits, leaving the asymptotic value poorly constrained. Appendix G.4 compares four candidate extrapolation forms and identifies the times at which their predictions separate.

This window dependence accounts for the entropy-route values in Table 1, which are fitted rather than fixed to $c = \frac{1}{2}$ as in Figure 8. The longer ladders at $p = 0$ and $p = 0.1$ recover the Ising value. The shorter ladders at $p = 0.3$ and $p = 0.5$ instead yield larger values, consistent with the direction of the finite-time correction but not yet converged. The truncation control at the integrable point quantifies the corresponding shift when the fit is restricted to a short window.

The accessible time window shortens as p increases, leaving fewer points with which to constrain the decay. We terminate each window when the ensembles described in Section 6.3 cease to produce physical entropy profiles reliably. The fraction of unphysical runs increases with T at every coupling, and almost no runs remain usable beyond the windows reported in Table 1. At $p = 0.5$ and $T = 6$, none of the runs produces a physical profile (Appendix F.6).

7.2 The spectral route

We compute the leading spectrum over the time windows listed in Table 1. As shown in Figure 9, the trajectory of μ_0 remains close to a circle whose radius depends on the coupling: its phase winds with T, while its modulus varies by roughly 1% or less over each ladder. This approximately constant modulus is the signature of emergent dual unitarity discussed in Section 2.2. The phase advance also provides a criterion for terminating each ladder. It is nearly constant along the physical branch, and a persistent departure from this behaviour indicates that the branch is no longer resolved.

We extract the central charge from the phase curvature using Eq. (6) and the fitting conventions of Appendix G.4. The resulting fits are shown in Figure 10.

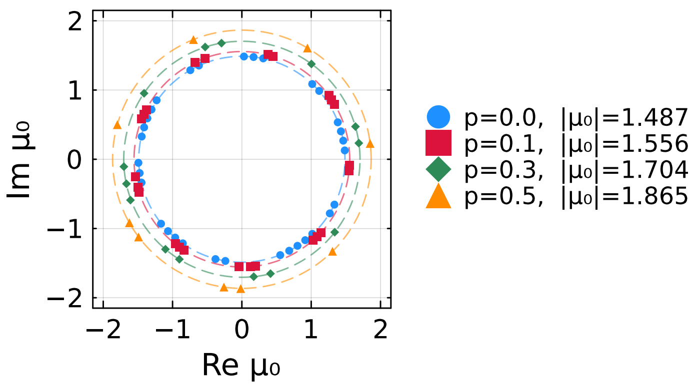


Figure 9: **Trajectory of the leading eigenvalue.** For each coupling, the dashed circle has the mean modulus of the corresponding ladder and is labelled by that value. The phase winds with T, while the modulus remains approximately constant.

The spectral estimates in Table 1 remain within approximately 10% of the Ising value at every coupling and are consistent with $c = \frac{1}{2}$ within the resolution of the fits. The equilibrium chord analysis of Section 4.1 gives comparable estimates over the same coupling range. This agreement provides an independent comparison between universal data extracted from the ground state and from real-time dynamics.

The two routes deviate from the conformal value for different reasons. In the entropy route, the plateau approaches $\pi c/16$ from above, and its asymptotic value must be obtained through the extrapolation described in Section 7.1. At the integrable point, the spectral estimate is instead biased downwards to approximately 0.45 by two finite-parameter corrections quantified separately in Appendix G.1. First, Eq. (6) gives only the leading term in an expansion in $1/T$; including the next-order term raises the fitted coefficient from 90% to 97% of its predicted value. Second, the regulator contributes through β_0/T. The fitted coefficient decreases linearly with β_0 and extrapolates to the predicted value as $\beta_0 \to 0$. That limit is an extrapolation rather than an accessible setting, since without the cooling interval the initial state is a bare product state and not a conformal boundary state (Section 2.1), so the strip would have no well-defined conformal edge from which to read the universal data, while the production value $\beta_0 = 0.2$ produces a downward shift of approximately 10%. The observed discrepancy is therefore consistent with controlled finite-time and finite-regulator effects.

The sound velocity affects the spectral analysis in two ways. Equation (7) gives phase gaps proportional to $1/(vT)$, so a larger velocity compresses the tower at fixed evolution time and requires finer phase resolution. The conversion from the fitted curvature to the central charge also carries a factor of the velocity, since $c = 24\,v\,|C|/\pi$, so a fixed uncertainty in C produces an uncertainty in c that grows with v. Both effects reduce the spectral resolution at stronger coupling.

The windows of Table 1 also shorten as p increases, and we do not attribute this to a single cause. Both effects above contribute, and the conditioning of the left and right states is measured to worsen sharply with the coupling for reasons that remain unresolved (Appendix F.6). The present measurements do not separate these contributions.

The Ising boundary spectrum persists after introducing the NNN interaction. Figure 11 shows the seven leading dimensions resolved with $k = 8$. The leading dimension, reported

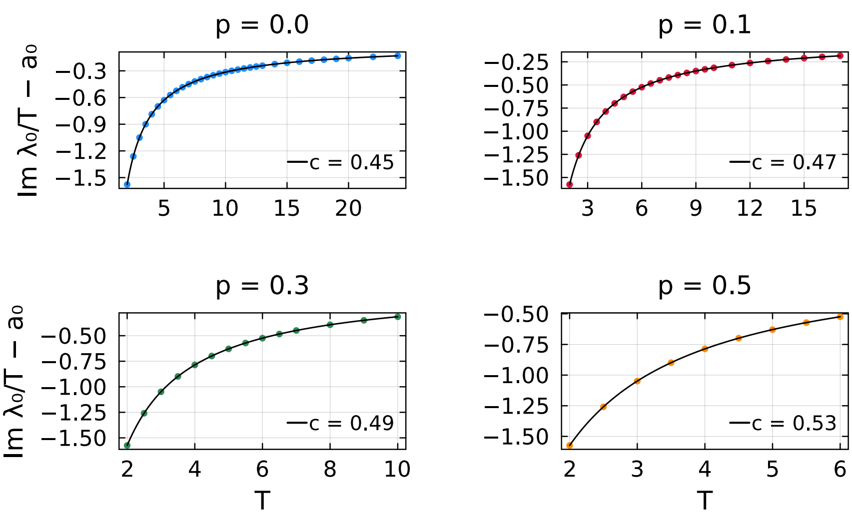


Figure 10: **Central charge from the phase of the leading eigenvalue.** Fits of Eq. (6) to $\operatorname{Im}\lambda_0/T$ at each coupling. The fitted constant a_0 is subtracted to display only the time-dependent contribution, whose curvature determines the central charge reported in each legend.

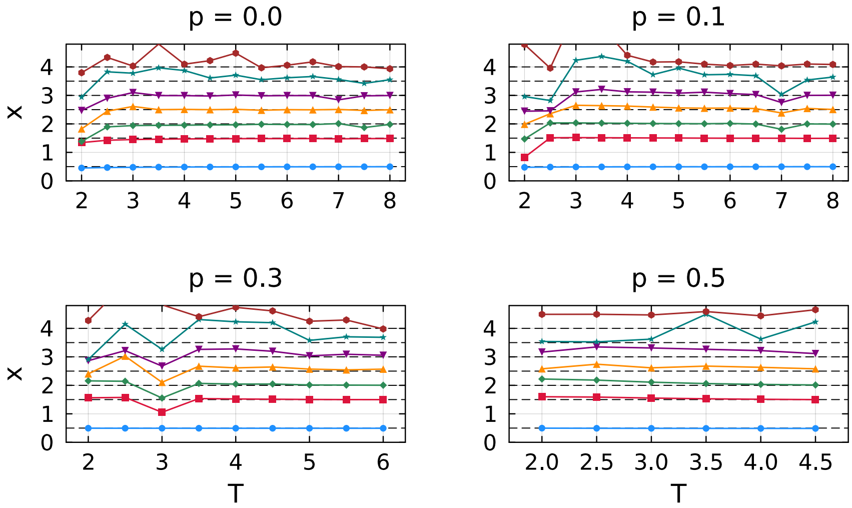


Figure 11: **Boundary dimensions as a function of evolution time.** The seven leading dimensions at each coupling are obtained from the phase gaps through Eq. (7), ordered by increasing phase distance from μ_0, and resolved using $k = 8$. The dashed lines mark the predicted values $\frac{1}{2}$, $\frac{3}{2}$, 2, $\frac{5}{2}$, 3, $\frac{7}{2}$, and 4. These values belong to the two towers selected by the free boundary condition: the identity tower contains the integers, while the energy tower contains the half-integers.

in Table 1, remains within 3% of the predicted value $\frac{1}{2}$ at every coupling.

The first five dimensions remain close to their predicted values throughout the coupling range. The sixth and seventh show a systematic upward drift, reaching 4.23 and 4.65 at $p = 0.5$, compared with the predicted values $\frac{7}{2}$ and 4. They are obtained from the highest states retained by the block and are therefore the least converged, as quantified directly in Appendix F.2. Their phase gaps are also the most difficult to resolve as the spectrum becomes increasingly compressed.

8 Conclusions and Outlook

This thesis asked whether the conformal signatures that the Loschmidt echo exposes in an integrable chain survive when integrability is broken. We studied a self-dual ANNNI-type chain at its critical point, mapped the echo to a CFT on a strip, and read the universal data from the spectrum of the temporal transfer matrix. For the couplings we can reach, $p \leq 0.5$, all four predictions of Section 2.2 are confirmed.

The leading eigenvalue keeps an almost constant modulus while its phase winds, which is the emergent dual unitarity of Eq. (5). The curvature of that phase gives a central charge consistent with $c = \frac{1}{2}$ within the resolution of the fits, and the generalised temporal entropy reproduces the predicted chord profile with an imaginary plateau extrapolating to the same value. The phase gaps give the boundary spectrum, where a block of eight temporal states resolves its first seven members at every coupling, following the two conformal towers that the free boundary condition selects. The NNN interaction therefore changes the non-universal scales without altering the universal content we can resolve.

Reaching that conclusion required work on several fronts. The model had to be placed in the Ising class independently of the dynamics, which DMRG on the ground state does up to $p = 1.5$, and the sound velocity had to be measured at each coupling by exact diagonalisation of finite rings. The velocity is not a check but an input, since every prediction converts a phase into universal data through v.

The evolution operator required its own construction. An NNN interaction cannot be assigned to a single bond of a brick-wall Trotter circuit, and the transverse contraction needs the one-site translation invariance that such a circuit destroys. We therefore represent the Hamiltonian as a finite-state machine and exponentiate that representation directly, using a second-order construction that reproduces every overlapping pair of Hamiltonian terms in a single application. We implemented it in an existing exponential-MPO library, and we extract the repeated bulk tensor from a five-site patch so that the channel carrying the NNN interaction survives into the temporal column.

The spectral machinery follows from what each observable needs. The entropies and the leading eigenvalue require only the dominant pair, which the single-vector iteration provides, while the boundary content requires the subleading eigenvalues and therefore the block iteration, whose implementation is given in Appendix F. The two differ sharply in cost. The single-vector iteration propagates one state on each side and is cheap enough to repeat from many seeds and to reach the longest times in this thesis, whereas the block iteration propagates eight and recombines them at every step, so the boundary content is resolved only up to $T = 8$, 6 and 4.5 at $p = 0.1$, 0.3 and 0.5. What can be quoted from either is decided by the numerical controls. We compared the two truncation schemes, varied the bond dimension and the singular-value cutoff, halved the time step, repeated every entropy point from independent random seeds, and tracked the physical branch by continuity rather than by largest modulus. These fix the windows of Table 1.

The point at which the method stops is what we understand least. The measurements establish that the eigenvalues are far more robust than the eigenvectors, since $|\mu_0|$ reproduces across independent seeds at every usable time while the entropies built from the same runs do not, and that the failures set in earlier as the coupling grows. The failure accompanies the left and right states becoming ill-conditioned, with their overlap falling. The controls of Appendix F.6 rule out the obvious explanations, since the effect survives exact diagonalisation without any truncation, it is not accounted for by the leading eigenvalues approaching one another, and no single numerical parameter removes it. What property of the NNN coupling controls the conditioning therefore remains open, and the windows we report characterise the reach of the present configuration rather than a limit of transverse contraction itself.

Several improvements would extend that reach. Understanding the conditioning would indicate in advance how far a given model can be pushed, and the sound velocity already provides part of that estimate, since the phase gaps scale as $1/(vT)$ and a velocity measured in equilibrium predicts how compressed the spectrum will be before any dynamical calculation is run. The block iteration also leaves room for improvement. Most of its cost lies in recombining the propagated states, and its stopping criterion does not account for the truncation noise floor, so a cheaper and better conditioned implementation would lengthen the accessible windows and make the identification of the physical branch among the Ritz values more reliable, which is the step that the non-normality of the transfer matrix makes delicate. With longer windows, $c(p)$ and $x_1(p)$ could be followed towards the edge of the critical phase, where the equilibrium results remain consistent with Ising criticality but the present dynamical windows are too short.

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A Conformal-field-theory derivations

Section 2 relates the Loschmidt echo of a critical quench to the partition function of a CFT on a strip whose width is determined by the evolution time. The analysis uses two results derived here: the logarithmic coefficient of the generalised temporal entropy and the spectrum of the transfer matrix along the strip. We first obtain the entropy coefficient and chord variable from twist-operator correlation functions, including the factor of two associated with an edge-anchored bipartition. We then quantise the theory on the strip and continue its width to complex time, obtaining Eqs. (2), (6), and (7).

A.1 Conformal maps, the two-point function and the entropy coefficients

In two dimensions, we write the Euclidean coordinates as $z = x + i\tau$ and $\bar{z} = x - i\tau$. A conformal map is a transformation $z \to w(z)$ depending on z alone, together with an independent transformation $\bar{z} \to \bar{w}(\bar{z})$ of the conjugate coordinate. The holomorphic and antiholomorphic dependences therefore transform separately, and we call each one a chiral sector; in real time they carry the left- and right-moving excitations. A primary operator ϕ is one that transforms under such a map with the Jacobian alone, without the derivative terms a general operator acquires, and its conformal weights $(h, \bar{h})$ set the exponent contributed by each sector. Their sum,

$$x = h + \bar{h} \tag{A.1}$$

is the scaling dimension used in the main text, and their difference $s = h - \bar{h}$ is the spin, which measures the response to a rotation about the operator. The twist operators used below sit at branch points of the replica surface, which are invariant under such a rotation, so $h = \bar{h}$ and $s = 0$. On the plane, the two-point function has the power-law form

$$\langle \phi(z_1)\phi(z_2)\rangle = \frac{1}{(z_1 - z_2)^{2h}\,(\bar{z}_1 - \bar{z}_2)^{2\bar{h}}}. \tag{A.2}$$

Under a conformal map $z \to w$, a primary field transforms with the corresponding Jacobian. Equation (A.2) therefore determines its correlator on any geometry conformally related to the plane [29]. The logarithmic map $w = (\beta/\pi)\log z$ maps the plane to a cylinder of circumference β and the upper half-plane to a strip of width β. Here β is a length, so on the lattice it carries the sound velocity through $\beta = v(2\beta_0 + iT)$, and v does not appear separately in the map. The transformed correlator is [1, 32]

$$\langle \phi(\omega_1)\phi(\omega_2)\rangle \;\propto\; \left[\sinh \tfrac{\pi(\omega_1-\omega_2)}{\beta}\right]^{-2h} \left[\sinh \tfrac{\pi(\bar{\omega}_1-\bar{\omega}_2)}{\beta}\right]^{-2\bar{h}}. \tag{A.3}$$

The separation on the plane is replaced by the chord distance $(\beta/\pi)\sinh[\pi(\omega_1 - \omega_2)/\beta]$ on the cylinder or strip. Analytic continuation of the width to real time replaces the hyperbolic sine by a sine. This continued chord distance gives the variable W in Eq. (4).

The entropy coefficients follow from the same correlation function. The Rényi entropy of a region $\mathcal{A}$ is

$$S_n = \frac{1}{1-n}\log \operatorname{Tr}\rho_{\mathcal{A}}^n, \tag{A.4}$$

so the quantity to be computed is the moment $\operatorname{Tr}\rho_{\mathcal{A}}^n$, the trace of the nth power of the reduced density matrix. For integer n this moment is itself a partition function: each factor of $\rho_{\mathcal{A}}$ is written as a path integral, and taking the trace joins n copies of the system cyclically along $\mathcal{A}$ [32, 41]. Analytic continuation to $n \to 1$ gives the von Neumann entropy. Each endpoint of $\mathcal{A}$ in the interior is a branch point of the replicated surface and is represented by a primary twist operator of the n-fold replicated theory with scaling dimension [32]

$$x_n = \frac{c}{12}\Big(n - \frac{1}{n}\Big), \tag{A.5}$$

whose weights are $h_n = \bar{h}_n = x_n/2$ in the notation of Eq. (A.1).

For an interval of length ℓ with both endpoints in the interior, the moments are determined by the two-point function of two twist operators. Equation (A.2) gives $\operatorname{Tr}\rho_{\mathcal{A}}^n \sim \ell^{-2x_n}$ and therefore

$$S_n = \frac{c}{6}\Big(1 + \frac{1}{n}\Big)\log\ell + \text{const}, \qquad S_{n\to 1} = \frac{c}{3}\log\ell + \text{const}. \tag{A.6}$$

The temporal bipartitions used in this thesis terminate at an edge of the strip and have only one endpoint in the interior. Their moments therefore involve the one-point function of a single twist operator. A conformal boundary relates the two chiral sectors, so the antiholomorphic part of an operator can be replaced by a holomorphic one at its mirror image [31]. The one-point function is then evaluated as a correlator between the operator and that image, separated by twice the distance to the boundary [32]. Because a single operator is inserted rather than two, the exponent is x_n instead of $2x_n$ and the logarithmic coefficient is halved:

$$S_n = \frac{c}{12}\Big(1 + \frac{1}{n}\Big)\log\ell + \text{const}. \tag{A.7}$$

On a strip of finite extent ℓ, the distance between an operator at x and its image is $(2\ell/\pi)\sin(\pi x/\ell)$, which is the argument of the chord variable $W(x,\ell)$ in Eq. (4), so the edge-anchored Rényi entropy is

$$S_n = \frac{c}{12}\Big(1 + \frac{1}{n}\Big)\,W(x,\ell) + s_0, \tag{A.8}$$

where s_0 is non-universal. For an open chain the same result is written with ℓ/π rather than $2\ell/\pi$ [32]. This difference contributes only an additive constant and is absorbed into s_0.

To obtain Eq. (3), we continue the strip extent to real time by setting $\ell \to i\,vT$ and $x \to i\,vt$. The ratio x/ℓ remains real, while the prefactor inside the logarithm acquires a factor of i. Using $\log i = i\pi/2$ gives

$$S_n^{\text{gen}}(t) = s_0 + \frac{c}{12}\Big(1 + \frac{1}{n}\Big)\Big[\,W(t,T) + \frac{i\pi}{2}\,\Big]. \tag{A.9}$$

This reproduces Eq. (3). The real part contains the chord profile, while the imaginary part is the constant $\frac{\pi c}{24}(1 + \frac{1}{n})$, equal to $\pi c/16$ at $n = 2$. The velocity contributes only an additional constant inside the logarithm and is absorbed into s_0.

A.2 The transfer-matrix spectrum from the stress tensor

The stress-energy tensor determines the response of the theory to coordinate transformations. In two dimensions, its modes L_n and $\bar{L}_n$ generate conformal transformations in the two chiral sectors and satisfy one copy of the Virasoro algebra each, whose central term is proportional to c [29]. The mode L_0 generates scale transformations, and its eigenvalues on primary operators are their conformal weights. Its spectrum therefore encodes the operator content.

Conformal boundary conditions at the two edges of the strip identify the chiral sectors. Extending the stress tensor into the lower half-plane through $\bar{T}(z) = T(z)$ leaves a single Virasoro algebra [31]. The dimensions below are therefore the weights h of boundary operators, rather than the bulk dimensions $h + \bar{h}$ defined in Eq. (A.1); we continue to write them x_i.

The generator of translations along the strip is obtained from the stress tensor. Since the stress tensor is not a primary field, a conformal map $z \to w$ introduces an inhomogeneous Schwarzian term proportional to the central charge [29]:

$$T(w) = \Big(\frac{\mathrm{d}z}{\mathrm{d}w}\Big)^2 T(z) + \frac{c}{12}\{z; w\}. \tag{A.10}$$

The map $w = (\beta/\pi)\log z$ from Appendix A.1 takes the upper half-plane to a strip of width β and has Schwarzian derivative $\{z; w\} = -\frac{1}{2}(\pi/\beta)^2$. Since $\langle T\rangle$ vanishes on the half-plane, its expectation value on the strip is $\langle T(w)\rangle = -\pi^2 c/(24\beta^2)$. Integrating across the strip gives a universal contribution $-\pi c/(24\beta)$ to the free energy per unit length [30]. Including the excitation levels generated by dilatations, the translation generator is [35]

$$\hat{H} = \frac{\pi}{\beta}\Big(L_0 - \frac{c}{24}\Big). \tag{A.11}$$

It contains both universal quantities of interest: the Casimir term $-c/24$ determines the central-charge contribution, while the spectrum of L_0 gives the boundary scaling dimensions. The transfer matrix advances the network by one spatial column, and $\hat{H}$ generates translations in that direction. Hence $\mathcal{E} = e^{-a\hat{H}}$, where a is the column spacing [1]. Substituting Eq. (A.11) and setting $a = 1$ in lattice units reproduces Eq. (2), with $g = a/\beta$ and eigenvalues

$$\mu_i = \exp\Big[-\pi g\Big(x_i - \frac{c}{24}\Big)\Big]. \tag{A.12}$$

The identity sector has $x_0 = 0$ and gives the dominant eigenvalue. Differences between logarithmic eigenvalues therefore cancel the Casimir contribution and isolate the boundary dimensions.

The continuum expression must be supplemented by two non-universal lattice contributions. The free energy per unit length contains a bulk term and a contribution from the two boundaries [30], giving the level energies

$$E_i = f\beta + f_s + \frac{\pi}{\beta}\Big(x_i - \frac{c}{24}\Big), \tag{A.13}$$

where f and f_s depend on the microscopic model, Trotter step, and regulator. Since both terms are common to every level, they multiply all eigenvalues of $\mathcal{E}$ by the same factor. They therefore cancel in the phase differences used to extract the dimensions. The leading phase is instead fitted as a series in T. Since β is proportional to T, the bulk term $f\beta$ contributes a term linear in T, while f_s is real and shifts only the modulus. Neither reaches the $1/T$ coefficient, which is the one carrying the central charge, and neither contributes a constant to the phase either: $f\beta$ is linear in T and f_s is real. The constant term in $\mathrm{Im}\,\lambda_0$ is therefore the branch offset alone.

The effective strip width contains both imaginary- and real-time contributions. The regulator supplies imaginary-time intervals at the two boundaries, the quench supplies the intervening real-time interval, and the velocity converts these durations into lengths:

$$\beta = v\,(2\beta_0 + i\,T), \qquad \frac{1}{\beta} \simeq \frac{1}{v}\Big(\frac{2\beta_0}{T^2} - \frac{i}{T}\Big), \tag{A.14}$$

where the second expression is expanded to first order in β_0/T. Substitution into Eq. (2) gives

$$\log\mu_i \;\simeq\; \frac{i\pi}{v\,T}\Big(x_i-\frac{c}{24}\Big)\;-\;\frac{2\pi\beta_0}{v\,T^2}\Big(x_i-\frac{c}{24}\Big). \tag{A.15}$$

The universal contributions appear in the phases at order $1/T$, whereas the regulator changes the moduli at order β_0/T^2. The eigenvalues follow from the level energies as $\log\mu_i = -E_i$ at $a=1$. Restoring the non-universal terms of Eq. (A.13), setting $x_0=0$ for the identity, and taking the imaginary part of $\lambda_0 = \log(-\mu_0)$ gives

$$\operatorname{Im}\lambda_0 = A\,T + B + \frac{C}{T} + \mathcal{O}(T^{-3}),\qquad A=-fv,\quad B=-\pi,\quad C=-\frac{\pi c}{24\,v}. \tag{A.16}$$

This is the form of Eq. (6) fitted in Section 7. The extensive term $f\beta$ fixes A and the factor -1 in the logarithm fixes B, while the coefficient C contains the central charge. Taking the difference between two logarithmic eigenvalues removes the common terms, including the Casimir contribution, and gives Eq. (7). The same expansion gives the emergent dual unitarity of Eq. (5). Writing the width as a geometric series,

$$\frac{1}{\beta} = -\frac{i}{v\,T}\sum_{k\geq 0}\Big(\frac{2i\beta_0}{T}\Big)^k, \tag{A.17}$$

the prefactor is imaginary and each further factor multiplies by i, so the terms with even k are imaginary and those with odd k are real. Only the real part contributes to the moduli, which therefore receive even powers of $1/T$ alone, each carrying at least one power of β_0. There is consequently no $1/T$ term: that contribution is purely imaginary and goes entirely into the phase. Keeping the leading real term,

$$|\mu_i| = \exp\Big[-\frac{2\pi\beta_0}{v\,T^2}\Big(x_i-\frac{c}{24}\Big)\Big]. \tag{A.18}$$

The Casimir term is common to every level, as are the non-universal contributions of Eq. (A.13), since $\operatorname{Re}(f\beta) = 2fv\beta_0$ does not depend on the level. All of them cancel in the ratio, leaving

$$\frac{|\mu_i|}{|\mu_0|} = \exp\Big[-\frac{2\pi\beta_0\,(x_i-x_0)}{v\,T^2}\Big] \simeq 1-\frac{2\pi\beta_0\,(x_i-x_0)}{v\,T^2}, \tag{A.19}$$

which is Eq. (5) with $b_i = 2\pi\beta_0(x_i-x_0)/v$. The ratio therefore approaches unity both as T grows and as the regulator is removed at fixed T, and the emergent dual unitarity tested in Section 7 follows.

B Truncation schemes

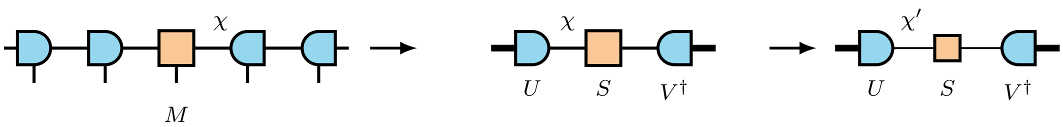
Truncation keeps the bond dimension bounded during transverse contraction. We now examine the two schemes introduced in Section 3.4, based on the RDM and the RTM. We first explain why Schmidt-value truncation is optimal for a single state. We then use gauge transformations to distinguish physical spectra from representation-dependent ones and identify the approximations involved in the RTM scheme.

Consider a cut that divides the chain into a block $\mathcal{A}$ and its complement $\mathcal{B}$ [3, 40]. Grouping the physical indices on each side into a composite index represents the wavefunction as a matrix. Its Singular Value Decomposition (SVD) gives the Schmidt decomposition

$$|\psi\rangle = \sum_{\alpha=1}^{\chi} s_\alpha\;|\alpha\rangle_{\mathcal{A}}\otimes|\alpha\rangle_{\mathcal{B}}\,,\qquad s_\alpha\geq 0,\quad \sum_\alpha s_\alpha^2 = 1, \tag{B.1}$$

where the states on each side form orthonormal bases. In mixed canonical form, the Schmidt coefficients appear as a diagonal matrix on the cut bond. The corresponding RDM is diagonal in the same basis, with eigenvalues $p_\alpha = s_\alpha^2$ that determine the entanglement across the cut. Reducing the bond dimension from χ to $\chi' < \chi$, as shown in Figure 12, retains the χ' largest Schmidt coefficients. The Eckart–Young–Mirsky theorem [53, 54] establishes that this is the closest state to $|\psi\rangle$ among all states of Schmidt rank at most χ'. Its squared norm error is the discarded weight,

$$\varepsilon = \| \, |\psi\rangle - \left|\tilde{\psi}\right\rangle \|_2^2 = \sum_{\alpha > \chi'} s_\alpha^2. \tag{B.2}$$



(a) canonical form

(b) we group the legs to form matrices

(c) we truncate

Figure 12: **Schmidt truncation at a bond.** (a) Chain in mixed canonical form, with the orthogonality centre M at the cut. (b) Grouping the indices on either side into composite indices (thick lines) gives the matrix decomposition $M = USV^\dagger$. (c) Truncated decomposition, with the internal dimension reduced from χ to χ'.

B.1 The RDM scheme

In canonical form, the RDM scheme obtains this truncation from a local tensor, as shown in Figure 13.

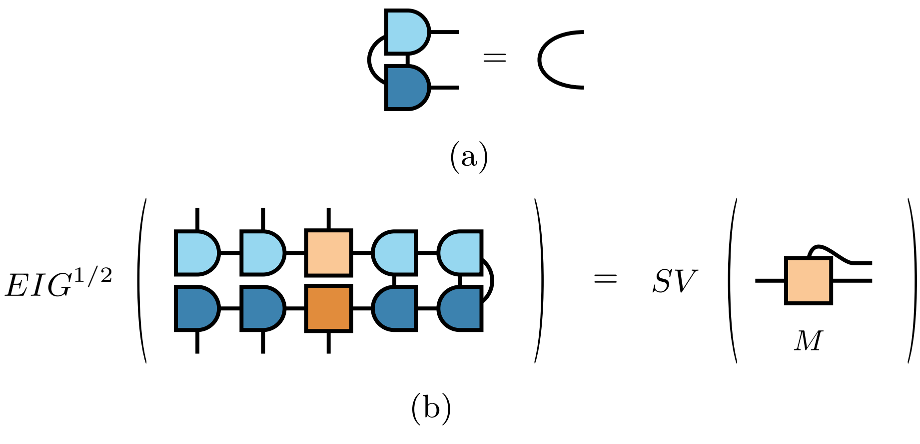


Figure 13: **Local RDM truncation in canonical form.** (a) Contracting a canonical tensor with its conjugate over the physical index and one virtual index gives the identity on the remaining index. Lighter and darker shades distinguish the tensor and its conjugate. (b) Reduced density matrix at a cut, with the left block open and the right block traced. The surrounding isometries contract to identities, so the eigenvalues of the full RDM are the squared singular values of the centre tensor M.

The truncation data can therefore be extracted from a local $\chi \times \chi$ environment matrix M_A rather than from the exponentially large global object. For a bipartite cut,

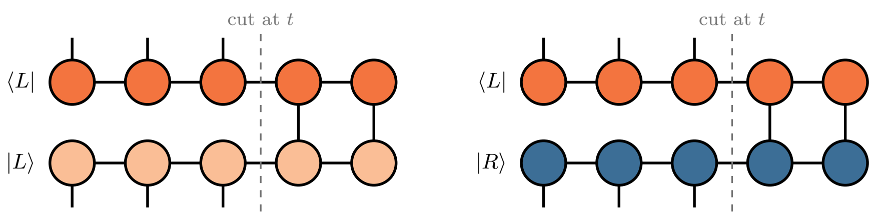
$$\rho_B = U_B \, M_A \, U_B^\dagger, \tag{B.3}$$

where U_B is an isometry because it is constructed from left-orthogonal tensors, so $U_B^\dagger U_B = \mathbb{1}$. Equation (B.3) therefore preserves the nonzero spectrum, $\mathrm{eig}(\rho_B) = \mathrm{eig}(M_A)$ up to additional zeros. The local environment carries the same eigenvalues as the full RDM.

This construction relies on the RDM being Hermitian and positive. Its eigenvalues are the squared Schmidt coefficients and can therefore be ordered unambiguously, while the Eckart–Young–Mirsky theorem provides a variational error bound. Neither property extends directly to a non-Hermitian transition matrix.

B.2 The RTM scheme and its limits

In the transverse picture, the transfer matrix is non-Hermitian and its dominant left and right fixed points are distinct. The corresponding truncation object is the RTM in Eq. (11), constructed from both states. Its eigenvalues can be complex and do not have the Schmidt interpretation described above. Moreover, a gauge transformation can redistribute weight between components that contribute to $\langle L|R\rangle$ and those that do not [39]; the magnitude of a component in either state separately is therefore not a gauge-invariant measure of its contribution to the overlap. Figure 14 compares the RDM and RTM at the same temporal cut.



(a) $\rho_L = \mathrm{Tr}_{T-t} |L\rangle \langle L|$

(b) $\mathcal{T}_t = \mathrm{Tr}_{T-t} |R\rangle \langle L| \,/\, \langle L|R\rangle$

Figure 14: **Possible truncation objects at a temporal cut.** Sites after the cut are traced over, while the physical legs before the cut remain open and form the matrix indices. (a) The RDM pairs a temporal state with its conjugate and is Hermitian and positive. (b) The RTM pairs the distinct fixed points $|R\rangle$ and $\langle L|$; it is generally non-Hermitian and can have complex eigenvalues.

The tensor representation of either object is not unique. Let $A^{[n]}$ and $A^{[n+1]}$ be the MPS tensors on two sites joined by a virtual bond. We may insert $\mathbb{1} = XX^{-1}$ on this bond and absorb one factor into each tensor,

$$A^{[n]} \;\to\; A^{[n]}X, \qquad A^{[n+1]} \;\to\; X^{-1}A^{[n+1]}, \tag{B.4}$$

without changing the represented state, because the two factors cancel when the bond is contracted [3, 4]. The physical overlap is invariant under this transformation,

$$\langle L|R\rangle \;=\; \langle L|\, XX^{-1} \,|R\rangle \;=\; \left\langle \tilde{L} \middle| \tilde{R} \right\rangle, \tag{B.5}$$

but the spectra used for truncation need not be. Their transformation depends on whether the state across the cut is paired with its conjugate or with an independent left state.

The RDM pairs the state with its conjugate. Conjugating the first transformation in Eq. (B.4) gives $A^{[n]*} \to A^{[n]*}X^*$. Thus, where the ket carries X, the bra carries $X^\dagger$, and

$$\rho_L \;\to\; X^\dagger \rho_L X. \tag{B.6}$$

The factors do not cancel unless X is unitary, because $X^\dagger X \neq \mathbb{1}$ in general. Equation (B.6) is a congruence transformation and does not preserve the eigenvalues. A gauge transformation can therefore move components that contribute to the overlap into the tail of the isolated RDM spectrum, where they may be discarded by truncation.

The RTM instead pairs $|R\rangle$ with the independent state $\langle L|$. The left tensors absorb X^{-1} on the other side of the same bond, following the second transformation in Eq. (B.4). Consequently,

$$\mathcal{T}_t \;\rightarrow\; X^{-1}\mathcal{T}_t X. \tag{B.7}$$

The factors now cancel because $X^{-1}X = \mathbb{1}$. Equation (B.7) is a similarity transformation for any invertible X and therefore preserves the eigenvalues. Figure 15 shows the corresponding network contractions for both schemes.

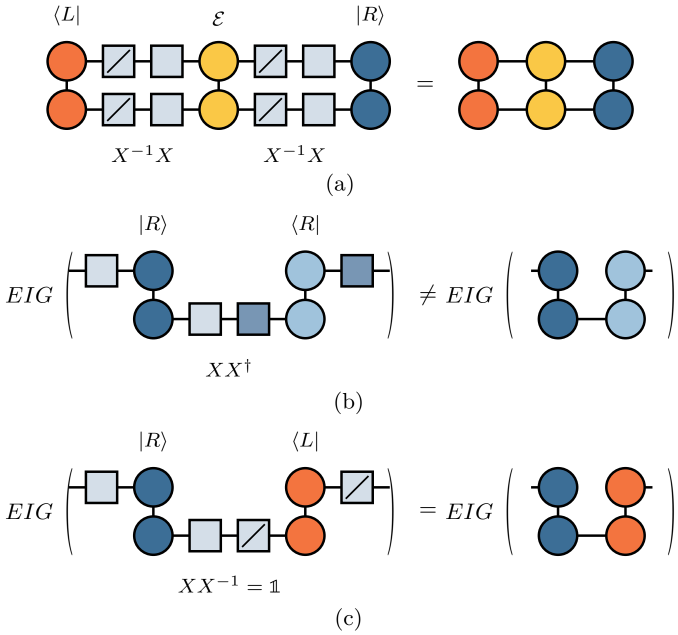


Figure 15: **Gauge transformation of the RDM and RTM.** (a) Inserting $X^{-1}X$ on a virtual bond changes the tensors but not the full contraction. In (b) and (c), the lower time site is contracted while the upper site remains open. (b) For the RDM, the gauge factors combine as $XX^\dagger \neq \mathbb{1}$ and produce the congruence $X\rho X^\dagger$, which changes the spectrum. (c) For the RTM, they combine as $XX^{-1} = \mathbb{1}$ and produce the similarity transformation $X\mathcal{T}_t X^{-1}$, which preserves the spectrum.

The spectrum of a single temporal-state RDM therefore depends on the chosen representation, whereas the RTM spectrum is invariant for the pair $\langle L|, |R\rangle$. Fixing a canonical gauge makes the RDM truncation well defined, allowing it to be used as a complementary numerical scheme despite this representation dependence.

On the other hand, the local reduction of the RTM differs significantly from that of the RDM. For two distinct states, the construction leading to Eq. (B.3) instead gives

$$\tau_B = U_B\, M_A\, V_B, \qquad U_B \neq V_B^\dagger, \tag{B.8}$$

which is not a similarity transformation. Its eigenvalues are therefore not preserved: $\mathrm{eig}(\tau_B) \neq \mathrm{eig}(M_A)$. However, U_B and V_B remain isometries and preserve singular values,

$$\mathrm{sv}(\tau_B) = \mathrm{sv}(M_A). \tag{B.9}$$

Thus, the local environment and the full RTM have the same singular values. These are the invariant quantities available from the local reduction and provide the truncation criterion we use in practice.

This local reduction requires a canonical form adapted to the left–right pair. As shown in Figure 16, the canonical condition is imposed on the pair $(\langle L| , |R\rangle)$ rather than on a state and its conjugate. Tensors away from the cut then contract to identities, leaving a local decomposition with the singular values of the full RTM. The two schemes consequently have the same local structure and comparable cost at fixed bond dimension.

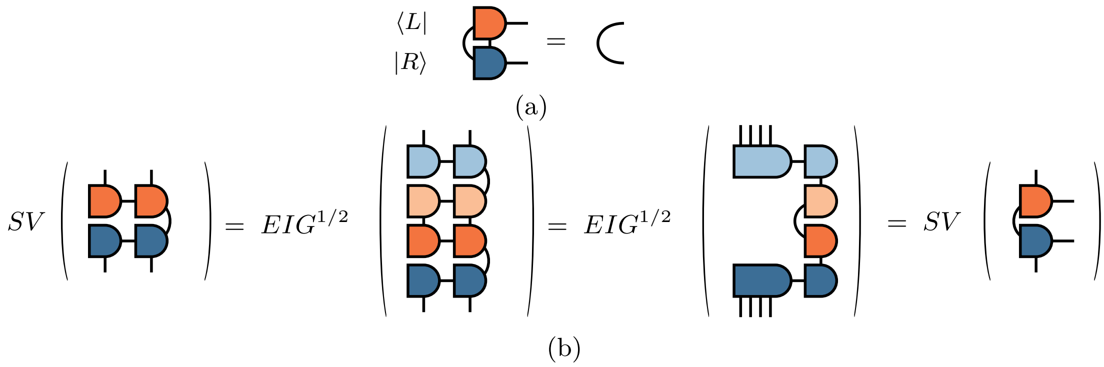


Figure 16: **Local RTM truncation in generalised canonical form.** (a) The contraction pairs a tensor of $\langle L|$ with the corresponding tensor of $|R\rangle$ rather than with its conjugate. (b) Imposing this condition along the chain reduces the environment to the tensors at the cut, whose singular values equal those of the RTM. The construction remains local but no longer has a variational error bound.

The potential reduction in bond dimension arises because the RTM measures relevance through the left–right overlap. The method uses a biorthogonal decomposition of $|R\rangle \langle L|$ and retains the components with largest-magnitude eigenvalues [18]. For the transfer matrices considered here, this can reach a given accuracy with a smaller bond dimension than the RDM scheme. The required RTM rank is bounded above by the operator entanglement of the bipartition [18].

The local truncation criterion is expressed through singular values, whereas the physical overlap is determined by the trace of $\mathcal{T}_t$. These quantities are not equivalent. Singular values satisfy $\mathrm{sv}(M) = \sqrt{\mathrm{eig}(MM^\dagger)}$ and coincide with the eigenvalues for a Hermitian positive RDM, but not for a non-Hermitian RTM. The trace $\mathrm{Tr}\,\mathcal{T}_t = \sum_i \lambda_i$ is a signed sum and is not controlled directly by singular-value magnitudes. For example, eigenvalues $+1$ and -1 both have unit magnitude but cancel in the trace.

The RTM truncation therefore lacks the variational bound available for an RDM, and the same criterion need not be reliable for a generic non-Hermitian matrix, whose eigenvalues can respond singularly to perturbations when the matrix is close to defective [55]. The error in the overlap can nevertheless be bounded: if the truncation changes the left and right temporal states by at most ϵ_L and ϵ_R in norm, then the error in $\langle L|R\rangle$ is at most $\epsilon_L + \epsilon_R + \epsilon_L\epsilon_R$ [16]. The RTM truncation targets contributions to the overlap rather than minimising these norm errors, so the discarded singular values cannot be used directly to evaluate this bound. Cancellation would require retained and discarded components of comparable size, which the quickly decaying spectra of the model studied here at the critical point do not provide. The different guarantees are reflected in the convergence: the RDM scheme generally converges smoothly, whereas the RTM result can vary irregularly.

Both schemes make one further approximation. At each stage of the sweep, the truncation uses the current local environment without including the uncontracted remainder of the network. Translation invariance reduces the resulting error because every subsequent bulk column applies the same operator and can regenerate discarded components of the leading subspace.

C Why the NNN coupling breaks integrability

Mapping a one-dimensional spin chain to fermions provides a direct test of whether it reduces to free fermions. Under the JW transformation, a quadratic fermionic Hamiltonian can be reduced to independent normal modes and diagonalised exactly. Terms of fourth order instead couple these modes and remove this free-fermion integrability. We apply this criterion to Eq. (13).

The mapping must account for the different algebras of spins and fermions: Pauli operators on different sites commute, whereas fermionic operators anticommute. The JW transformation achieves this through a parity string over all sites to the left of the operator. In the convention of Eq. (13), with the transverse field along σ^x and the order parameter along σ^z, it is [37, 56]

$$\sigma_j^x = 1 - 2n_j, \qquad \sigma_j^z = (c_j + c_j^\dagger)\, K_j, \qquad K_j = \prod_{k<j} (1 - 2n_k), \tag{C.1}$$

where c_j annihilates a spinless fermion, $n_j = c_j^\dagger c_j$, and K_j measures the fermion parity to the left of site j.

Two identities determine how these strings cancel. Each parity factor squares to the identity, $(1-2n_k)^2 = 1$, and hence $K_j^2 = 1$. Moreover, $1 - 2n_k = (-1)^{n_k}$ is even in the fermionic operators and commutes with operators on other sites, while on the same site

$$(c + c^\dagger)(1 - 2n) = c^\dagger - c. \tag{C.2}$$

The transverse field maps directly to the chemical-potential term $\sigma_j^x = 1 - 2c_j^\dagger c_j$. For the NN interaction, using $K_{j+1} = K_j(1 - 2n_j)$ gives

$$\begin{aligned}
\sigma_j^z \sigma_{j+1}^z &= (c_j + c_j^\dagger) K_j\, (c_{j+1} + c_{j+1}^\dagger) K_j (1 - 2n_j) \\
&= (c_j + c_j^\dagger)(c_{j+1} + c_{j+1}^\dagger)\, K_j^2\, (1 - 2n_j) \\
&= (c_j + c_j^\dagger)(1 - 2n_j)\, (c_{j+1} + c_{j+1}^\dagger) \\
&= (c_j^\dagger - c_j)(c_{j+1} + c_{j+1}^\dagger).
\end{aligned} \tag{C.3}$$

The string commutes with the operators on site $j+1$ because it is even in the fermions. The remaining factor $(1-2n_j)$ is then combined with the operator on site j using Eq. (C.2). No parity factor survives, so the interaction is quadratic. Together with the field term, this makes the $p = 0$ chain a free-fermion model that can be diagonalised exactly by a Bogoliubov transformation [45]. At criticality, its continuum limit is the free Majorana CFT with $c = \frac{1}{2}$ [29].

For the NNN interaction, the parity string crosses the intervening site. Since $K_{i+2} = K_i(1 - 2n_i)(1 - 2n_{i+1})$, the cancellation leaves two parity factors:

$$\begin{aligned}
\sigma_i^z \sigma_{i+2}^z &= (c_i + c_i^\dagger) K_i\, (c_{i+2} + c_{i+2}^\dagger) K_i (1 - 2n_i)(1 - 2n_{i+1}) \\
&= (c_i + c_i^\dagger)(1 - 2n_i)\, (1 - 2n_{i+1})\, (c_{i+2} + c_{i+2}^\dagger) \\
&= (c_i^\dagger - c_i)\, (1 - 2n_{i+1})\, (c_{i+2} + c_{i+2}^\dagger).
\end{aligned} \tag{C.4}$$

The factor on site i is absorbed as in the NN case, but the factor on site $i+1$ remains because the bond contains no spin operator there. The hopping and pairing between sites i and $i+2$ therefore depend on the occupation of the intervening site.

Expanding the remaining parity factor separates the quadratic and quartic contributions:

$$\sigma_i^z \sigma_{i+2}^z = (c_i^\dagger - c_i)(c_{i+2} + c_{i+2}^\dagger) - 2\,(c_i^\dagger - c_i)\, c_{i+1}^\dagger c_{i+1}\,(c_{i+2} + c_{i+2}^\dagger). \tag{C.5}$$

The second term contains four fermionic operators. The self-dual interaction likewise contains a density–density contribution,

$$\sigma_j^x \sigma_{j+1}^x = 1 - 2n_j - 2n_{j+1} + 4\, c_j^\dagger c_j c_{j+1}^\dagger c_{j+1}. \tag{C.6}$$

Both terms proportional to p therefore contain four-fermion interactions. These interactions couple the normal modes of the $p = 0$ chain, so Wick's theorem no longer reduces correlation functions to products of two-point functions [57], and no single-particle transformation diagonalises the Hamiltonian, since a linear transformation of the modes preserves the order of the interaction terms. Thus $p \neq 0$ removes the quadratic free-fermion structure responsible for integrability at $p = 0$.

D Equilibrium measurements

The dynamical analysis in Section 7 requires two equilibrium quantities. The equilibrium central charge provides an independent reference for the dynamical estimate, while the sound velocity converts transfer-matrix phase gaps into scaling dimensions. We determine both quantities here.

D.1 Finite-size scaling of the mid-chain entropy

Section 4.1 extracts the central charge from the entanglement profile at a fixed system size. We obtain an independent estimate from the size dependence of the entropy across the mid-chain cut. At this cut, the chord variable is maximal, $W(L/2, L) = \ln(2L/\pi)$, and Eq. (14) becomes

$$S_{\mathrm{vN}}(L/2) = \frac{c}{6}\,\ln L + k'. \tag{D.1}$$

A linear fit against $\ln L$ therefore has slope $c/6$. For $N = 40$–200, the uncorrected fit gives $c = 0.511$ at $p = 0.1$ and $c = 0.514$ at $p = 0.5$. The same ground states give the Rényi-2 entropy used in the temporal calculation, whose logarithmic coefficient is $c/8$. The corresponding uncorrected estimates, $c = 0.584$ and 0.588, lie appreciably above the Ising value. Figure 17(a) shows both estimators.

This discrepancy is caused by finite-size corrections associated with the conical singularity of the replica geometry. Operators that are relevant at this singularity produce corrections proportional to $\ell^{-x/n}$ for an interval anchored at the boundary, where x is the scaling dimension of that operator [42]. The mid-chain cut of an open chain has this geometry, so we fit

$$S_n(L/2) = \frac{c}{12}\Big(1 + \frac{1}{n}\Big)\ln L + k_n + A_n L^{-x/n}. \tag{D.2}$$

with A_n a non-universal amplitude. Which operator appears at the singularity is not known in advance, so we determine its dimension from the data: fits with x free return values close to 1 at both n, although a three-parameter fit to nine sizes does not determine it sharply. We therefore fix $x = 1$. The corrected estimates are $c = 0.500$ at both couplings for the von Neumann entropy and $c = 0.508$ and 0.505 for the Rényi-2 entropy, and the largest residual falls by two orders of magnitude in every case. Figure 17(b) shows the corrections that the fits absorb.

The factor x/n explains why the uncorrected estimates behave differently. For the same x, the correction to the von Neumann entropy decays as L^{-x}, whereas that of the Rényi-2 entropy decays as $L^{-x/2}$. The former is therefore already within 3% of $\frac{1}{2}$, while the latter requires the correction term. The analogous temporal correction scales as $\Delta S_n \propto T^{-1/n}$ [23] and is included in Section 7.1.

The finite-size calculations use fifteen DMRG sweeps, a bond-dimension ramp up to 400, and a truncation cutoff of 10^{-10}, beginning with noise-assisted sweeps. The $N = 400$ chord fit in Section 4.1 instead uses fifty sweeps, a maximum bond dimension of 1000, and a cutoff of 10^{-12}.

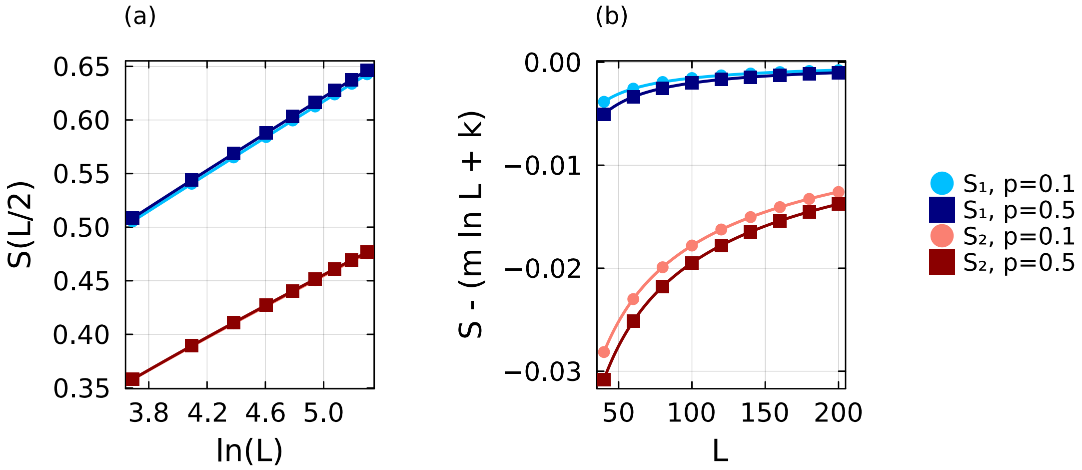


Figure 17: **Finite-size scaling of the mid-chain entropy.** (a) The von Neumann entropy S_1 and Rényi-2 entropy S_2 against $\ln L$ at $p = 0.1$ and $p = 0.5$, with the fits of Eq. (D.2). The corrected estimates are $c = 0.500$ at both couplings from S_1, and $c = 0.508$ and 0.505 from S_2. The correction changes the curves by less than 10^{-3} and is not visible on this scale. (b) The same data after subtracting the fitted logarithmic contribution, which isolates the finite-size correction.

D.2 The sound velocity from the equilibrium dispersion

We obtain the sound velocities used in Section 4.2 from the low-momentum dispersion, $\omega = v|k|$. On a periodic chain of N sites, the allowed momenta are $k_m = 2\pi m/N$, so a finite-size estimate follows from the lowest excitations at $m = 0$ and $m = 1$. These states must be identified within the same symmetry sector before their energy difference can be interpreted as a dispersion.

We therefore diagonalise the Hamiltonian in Eq. (13) at $\lambda = 1$ together with its translation symmetry. The one-site translation operator T has eigenvalues $e^{2\pi i m/N}$ and determines the momentum index m. It commutes with the Hamiltonian, so every level carries a definite momentum.

Degenerate eigenspaces require additional care because numerical diagonalisation returns arbitrary linear combinations of states with the same energy. We group levels whose energies differ by less than 10^{-8} and, within each cluster, diagonalise T, which restores the labels. A shift by N sites returns the ring to itself, so $T^N = \mathbb{1}$ and the eigenvalues of T are the Nth roots of unity, $e^{2\pi i m/N}$ with m an integer. An eigenstate therefore has

$\langle T \rangle = e^{2\pi i m/N}$, and its phase gives m directly. The computed phase is not exactly a multiple of $2\pi/N$, so we round it to the integer it must be, and assign

$$m = \mathrm{round}\left[\frac{N}{2\pi}\arg\langle T\rangle\right]. \tag{D.3}$$

With arg taken in $(-\pi, \pi]$ the index is signed, and the momentum is $k_m = 2\pi m/N$. Only the ten lowest levels are computed.

The finite-size velocity is then

$$v(N) = \frac{E_1(2\pi/N) - E_1(0)}{2\pi/N} = \frac{N}{2\pi}\left[E_1(2\pi/N) - E_1(0)\right]. \tag{D.4}$$

This is the discrete form of the low-momentum slope $\mathrm{d}\omega/\mathrm{d}k$, and is the estimator introduced for these chains [25]. In practice we take the lowest excitation at the smallest non-zero $|m|$ returned by the diagonalisation and divide by $|m|$; in every case examined this state has $|m| = 1$.

Nothing above uses the parity $P = \prod_i \sigma_i^x$, although it commutes with the Hamiltonian too. Ignoring it is safe only if the lowest states at $m = 0$ and $m = 1$ happen to share a parity, since otherwise their energy difference mixes two branches. We checked it directly. Labelling both symmetries at once means diagonalising $c_1 T + c_2 P$ inside each degenerate cluster, so that neither label spoils the other; recomputing from the odd-parity states alone then leaves the velocity unchanged to every digit we store, at $N = 10$ and $N = 12$ and for every coupling up to $p = 0.5$. The two states already share a parity, so we do not project.

We validate the conversion from a spectral gap to a scaling dimension using the anisotropic XY chain,

$$H_{\mathrm{XY}} = -\sum_j \left[\frac{1+\gamma}{2}\,\sigma_j^x\sigma_{j+1}^x + \frac{1-\gamma}{2}\,\sigma_j^y\sigma_{j+1}^y\right] - \lambda\sum_j \sigma_j^z, \tag{D.5}$$

written with the coupling on σ^x and the field on σ^z, the opposite convention to Eq. (13); at $\gamma = 1$ it reduces to the transverse-field Ising chain. The transformation of Appendix C maps it to free fermions with dispersion

$$\varepsilon(k) = 2\sqrt{(\lambda - \cos k)^2 + \gamma^2\sin^2 k}. \tag{D.6}$$

At $\lambda = 1$ the gap closes at $k = 0$, which is what puts the chain at criticality. Expanding there with $1 - \cos k \simeq k^2/2$ and $\sin k \simeq k$ gives $\varepsilon \simeq 2|k|\sqrt{\gamma^2 + k^2/4}$, so $\varepsilon \to 2\gamma|k|$ as $k \to 0$ and the velocity is exactly $v(\gamma) = 2\gamma$. At $\gamma = 1$ this returns the Ising value $v = 2$ used throughout. The leading boundary dimension remains $x_1 = \frac{1}{2}$. The test runs through Eq. (7), which gives the first gap as $\mathrm{Im}(\lambda_1 - \lambda_0) = \pi x_1/(vT)$. Its leading behaviour is thus $1/T$; the terms linear in T and constant in T are common to both eigenvalues and cancel in the difference, so the next correction is $1/T^3$. We therefore fit

$$\mathrm{Im}(\lambda_1 - \lambda_0) = \frac{a_1}{T} + \frac{a_3}{T^3}, \tag{D.7}$$

whose leading coefficient is $a_1 = \pi x_1/v$. Both quantities on the right are known: the initial product state realises the same free boundary as in Section 2.1, so $x_1 = \frac{1}{2}$, and $v = 2\gamma$ from the expansion of Eq. (D.6), which gives the exact value $a_1 = \pi/(4\gamma)$. Read the other way round, $x_1 = v\,a_1/\pi$ turns a measured slope into a dimension. As shown in Table 2, using the correct velocity gives $x_1 = 0.461$, 0.497, and 0.496. Including the coupling-dependent velocity therefore keeps the extracted dimension close to $\frac{1}{2}$.

γ	$v = 2\gamma$	a_1	$\pi/4\gamma$	x_1 from $v = 2\gamma$
0.5	1	1.449	1.571	0.461
1.0	2	0.781	0.785	0.497
1.5	3	0.520	0.524	0.496

Table 2: **Validation on the anisotropic XY chain.** The first spectral gap is fitted as $\mathrm{Im}(\lambda_1 - \lambda_0) = a_1/T + a_3/T^3$ over the converged times: $T = 2, 3, 4$ for $\gamma = 0.5$ and 1.0, and $T = 2, 4$ for $\gamma = 1.5$. The exact coefficient is $a_1 = \pi/(4\gamma)$ for $x_1 = \frac{1}{2}$. The final column converts the fitted coefficient using the correct velocity $v = 2\gamma$.

Two independent checks validate the finite-ring calculation. The Hamiltonian used here is constructed directly through bit operations, independently of the MPO, and their full spectra at $N = 8$ agree to within 7×10^{-14}. The second check turns our velocity into a central charge. For a periodic chain the ground-state energy per site is $f(L) = f_\infty - \pi c\, v/(6L^2)$ [30], and a finite difference between L and $L-1$ removes the unknown f_∞,

$$c = -\frac{6\,[\,f(L) - f(L-1)\,]}{\pi v\,\left(L^{-2} - (L-1)^{-2}\right)}. \tag{D.8}$$

Evaluated on our own ground-state energies and velocities at $N = 16$, this gives $c = 0.504$ at $p = 0$ and $c = 0.525$ at $p = 0.5$, consistent with previously reported values for this model [25].

The accessible ring sizes are far from the thermodynamic limit: at $p = 0$ the $N = 18$ estimate is 1.9899 against the exact value 2, and reaching three-digit accuracy directly would need about $N = 57$. We therefore extrapolate, and fix the form of the correction at the integrable point, where the answer is known. Writing the error as $2 - v(N) \propto N^{-\alpha}$, the rescaled quantity $[2 - v(N)]\,N^{\alpha}$ is constant to within half a per cent for $\alpha = 2$ alone, drifting downwards for $\alpha = 1$ and upwards for $\alpha = 3$. Extrapolating with those two powers misses the exact velocity by 2×10^{-2} and 6×10^{-3}, against 6×10^{-5} for $1/N^2$. A further D/N^4 term is not constrained by five sizes and does not improve the extrapolation.

We therefore fit

$$v(N) = v_\infty + \frac{C}{N^2} \tag{D.9}$$

at every coupling and use v_∞ throughout; Table 3 lists the data and fit parameters, and Figure 18 shows the extrapolations. What this buys is clearest where it can be checked: at $p = 0$ the extrapolated value lies within 6×10^{-5} of the exact result, against 1.0×10^{-2} for the raw $N = 18$ estimate. It also matters more as the coupling grows, since $|C|$ rises from 3.27 at $p = 0$ to 140.7 at $p = 1.5$.

Over the range studied the extrapolated velocity is close to linear in p, $v_\infty \simeq 2.02 + 6.42p$, which is the trend shown in Figure 5. The linearity is only approximate: the slope between successive couplings falls steadily from 6.70 near $p = 0$ to 5.35 between $p = 1.0$ and $p = 1.5$, so the growth flattens at stronger coupling, and a fit at fixed system size still drifts with N. We therefore treat the relation as a description of the trend, and every conversion in this thesis uses the separately extrapolated $v_\infty(p)$.

p	$N = 10$	$N = 12$	$N = 14$	$N = 16$	$N = 18$	v_∞	C
0.0	1.9673	1.9772	1.9833	1.9872	1.9899	1.99994	−3.27
0.1	2.5707	2.6007	2.6191	2.6312	2.6396	2.67016	−9.97
0.2	3.1565	3.2073	3.2387	3.2594	3.2738	3.32561	−16.96
0.3	3.7247	3.7972	3.8423	3.8722	3.8931	3.96723	−24.34
0.4	4.2756	4.3707	4.4305	4.4703	4.4981	4.59585	−32.16
0.5	4.8096	4.9284	5.0038	5.0543	5.0897	5.21214	−40.46
1.0	7.2534	7.5020	7.6674	7.7824	7.8653	8.12650	−88.23
1.5	9.4140	9.7946	10.0594	10.2504	10.3921	10.79924	−140.69

Table 3: **Finite-size velocities and thermodynamic extrapolations.** Values of $v(p)$ at $\lambda = 1$ for periodic chains of $N = 10$–18, with the parameters of the fit $v(N) = v_\infty + C/N^2$. The largest residual increases from 1.1×10^{-5} at $p = 0$ to 2.8×10^{-2} at $p = 1.5$. The conformal analysis uses the extrapolated values v_∞.

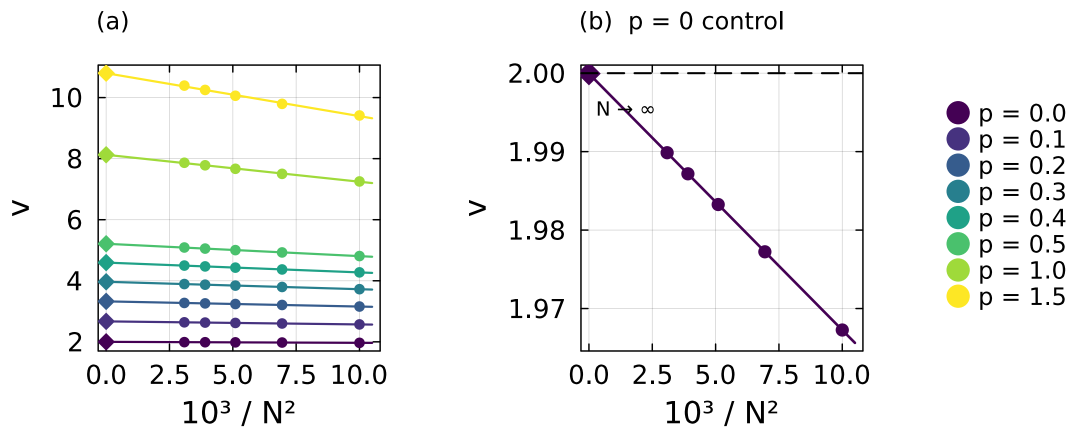


Figure 18: **Extrapolation of the sound velocity.** (a) Finite-size estimates of v against $1/N^2$ at each coupling, with fitted lines and thermodynamic intercepts (diamonds). (b) The integrable point on an expanded scale. The estimates are linear in $1/N^2$, and the intercept $v_\infty = 1.99994$ agrees with the exact value $v = 2$ (dashed line).

E Construction of the time-evolution MPO

This appendix describes the time-evolution MPO used in Section 5. We give its construction from the Hamiltonian MPO, the explicit local tensor of the second-order scheme, and numerical comparisons with the lower-order alternatives.

The construction begins from the Hamiltonian represented as an MPO. We supply the Hamiltonian as a symbolic sum of operator strings, from which the library assembles the FSM of Section 5.1. At each bond, it collects the terms that cross the bond, arranges their coefficients into a matrix, and compresses this matrix with an SVD. The retained components form the minimal set of memory channels required to carry the interactions across the bond. We denote their number by χ_H to distinguish it from the bond dimension χ used to truncate the temporal states. For the ANNNI-type Hamiltonian, the compression gives $\chi_H = 3$, corresponding to the three intermediate states in Eq. (15), and hence $D_W = \chi_H + 2 = 5$. It also determines the four blocks in that equation: D contains the on-site term, C initiates an interaction, B terminates it, and A propagates it across a site.

We construct the local evolution tensor with `ITensorExpMPO.jl` [58], which is built on `ITensors.jl` [59] and implements the operators W^{I} and W^{II} of Zaletel et al. [51]. Under the single-layer application considered here, neither operator is second-order accurate, as shown below. We therefore added the second-order cluster construction of Van Damme et al. [50]. The implementation is available in a fork of the package [60], together with its documentation.

E.1 The second-order *VD2* construction

We seek a local tensor $W(\tau)$, with $\tau = -i\delta t$, whose product along the chain approximates $e^{\tau H}$. Writing the Hamiltonian as $H = \sum_x H_x$, the expansion is

$$e^{\tau H} = \mathbb{1} + \tau \sum_x H_x + \frac{\tau^2}{2} \sum_{x,y} H_x H_y + \dots \tag{E.1}$$

Products of terms with disjoint support commute and factorise. A repeated local tensor therefore generates them without additional memory channels. The non-trivial contributions are products of overlapping terms. Since the corresponding blocks act on a common site, they need not commute, and both $H_x H_y$ and $H_y H_x$ must appear with the weights prescribed by Eq. (E.1). The accuracy of an evolution MPO is determined by which of these overlapping products it retains.

The constructions W^{I} and W^{II} encode the products through the virtual index [51]. Each memory channel is represented by an auxiliary hard-core boson whose occupation records whether a Hamiltonian term crosses the bond. Restricting the bond to at most one boson retains products in which no two terms cross the same bond. The more restrictive operator W^{I} keeps only disjoint-support products. The operator W^{II} also retains products that overlap on one site and consequently reproduces the on-site field to all orders. Both have virtual bond dimension $1 + \chi_H$, which is four for the present Hamiltonian.

Neither operator retains products in which two terms cross the same bond. This omission enters at order δt^2, making a single application first-order accurate for both the NNN model and the strictly NN Ising chain, as verified in Appendix E.2. Their errors nevertheless differ substantially. The operator W^{I} also omits products involving the on-site field, which occur throughout the chain, whereas W^{II} retains them. At $\delta t = 0.1$, this reduces the error by approximately a factor of five.

Second-order accuracy can be restored either by composing several stages [51] or by retaining the missing overlapping pairs in a single application. A staged composition is efficient in conventional time evolution because the sub-steps together advance one interval δt. In the transverse contraction, however, the stages must be blocked into a single repeated column. Two stages give a local physical dimension $4^2 = 16$ after rotation, compared with 13 for the second-order construction used here.

The two operators just described, and the one we use, all follow from a single move on the FSM of Section 5.1. In that machine the first level means that no interaction has started, the middle level that one is being carried across the bond, and the last level that one has been completed. To turn it into an evolution operator, the machine is made to return to the first level instead of stopping at the last one, picking up a factor τ on the way, so that the completed level can be dropped. Expanding the resulting tensor in powers of τ gives the identity, the full Hamiltonian, and then every disjoint product at every order with the weights of Eq. (E.1). This is how the first-order operators above are obtained. Two overlapping terms remain beyond them, since the machine carries only one interaction at a time.

The second-order operator follows from applying the same step to H^2 instead of H, as Figure 19 shows. Its machine has one level for every pair of levels of the original, which lets it tell apart the case where one term has acted and the other has not from the case where both act together. The first of these is a disjoint product, so those levels are returned to the starting level with a factor $\tau/2$. Once they are gone, the level recording two completed terms can only have been reached through terms that overlap, and returning it as well, now with $\tau^2/2$, supplies exactly the connected pairs that W^{I} and W^{II} omit. Van Damme et al. [50] give the general algorithm, its extension to any order, and the exact compressions that bring the result to the form written below.

Retaining clusters of up to n overlapping terms gives a one-step error of order δt^{n+1}. The first-order member is W^{II}, while the second-order member used here reproduces every connected pair. Representing two interactions that cross the same bond requires a doubled-memory sector of dimension χ_H^2, in addition to the scalar and single-memory sectors. The resulting virtual bond dimension is

$$1 + \chi_H + \chi_H^2. \tag{E.2}$$

For the ANNNI-type model, this gives 13, compared with 4 for W^{I} and W^{II}. At $p = 0$, only the NN interaction remains and $\chi_H = 1$, giving bond dimensions 3 and 2, respectively.

Before combining the blocks, we distribute the time step asymmetrically between the interaction endpoints. The initiating block carries $t_C = \sqrt{\delta t}$ and the terminating block carries $t_B = \tau/t_C$, so a completed interaction acquires one factor of τ:

$$\tilde{D} = \tau D, \qquad \tilde{C} = t_C\, C, \qquad \tilde{B} = t_B\, B, \qquad \tilde{A} = A. \tag{E.3}$$

This split is a rescaling of the virtual index. The two factors cancel when an interaction begins and ends within the chain, so physical matrix elements are unchanged. The rescaling instead balances the tensor entries at small δt, avoiding a representation in which one endpoint carries the complete time-step factor.

The local representation of $e^{\tau H}$ is written in terms of symmetrised products of the scaled blocks. We denote by $\{X_1 \dots X_m\}$ the sum over their distinct orderings, with $\{XY\} = X \otimes Y + Y \otimes X$, $\{XX\} = X \otimes X$, and $\{XXY\} = XXY + XYX + YXX$. Here $\otimes$ denotes the Kronecker product on the virtual memory index and ordinary operator multiplication on the physical index. The numerical prefactors below are the factorial weights from the exponential series. The expressions follow the second-order cluster operator of Van Damme et al. [50], specialised to the blocks of Eq. (15).

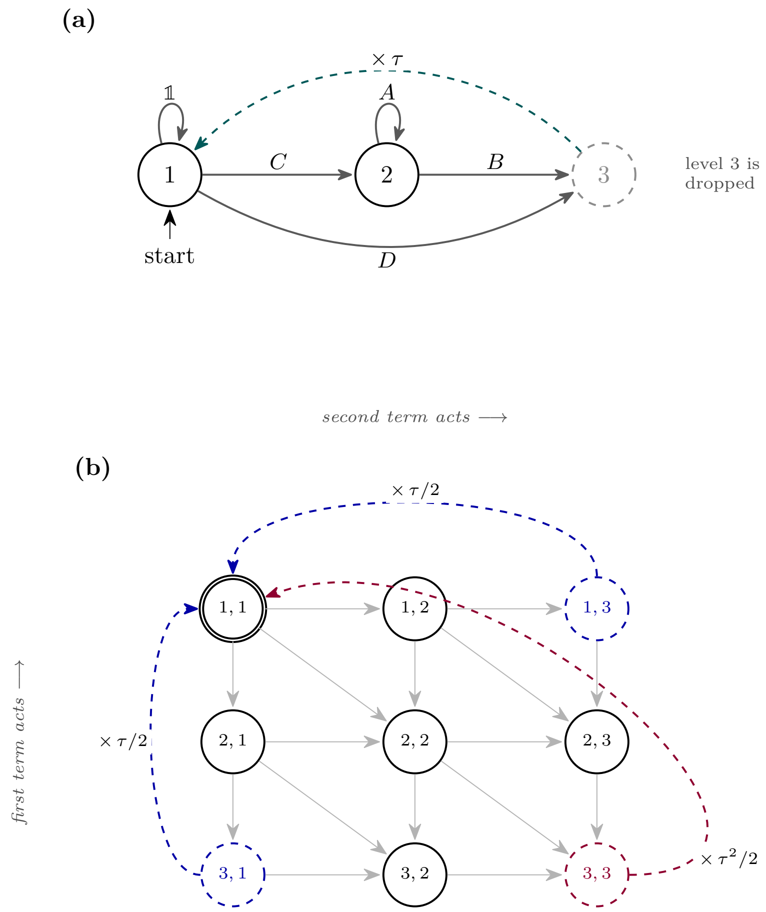


Figure 19: **Building the evolution operator from the Hamiltonian machine.** (a) The machine of Figure 6, reduced to its three levels. Making it return to level 1 instead of stopping at level 3, with a factor τ (teal), gives a first-order approximation of $e^{\tau H}$ and removes level 3 (grey). (b) The same move applied to H^2, whose levels are pairs, one index per Hamiltonian term. Grey arrows show the moves available, either index advancing on its own or both at the same site, while the jumps straight to level 3 made by the on-site term are left out to keep the picture readable. The disjoint pairs $(1, 3)$ and $(3, 1)$ fold back to the start with $\tau/2$ (blue), after which $(3, 3)$ holds only overlapping terms and folds back with $\tau^2/2$ (purple).

The local tensor has a 3×3 block structure of its own, not to be confused with the grid of Figure 19(b). There the two indices were the levels of the two Hamiltonian copies, whereas here the indices 1, 2, and 3 label the scalar, single-memory, and doubled-memory sectors that those levels reduce to once the equivalent ones are merged:

$$W^{VD2} = \begin{pmatrix} W_{11} & W_{12} & W_{13} \\ W_{21} & W_{22} & W_{23} \\ W_{31} & W_{32} & W_{33} \end{pmatrix}, \tag{E.4}$$

with

$$W_{11} = \mathbb{1} + \tilde{D} + \tfrac{1}{2}\{\tilde{D}\tilde{D}\} + \tfrac{1}{6}\{\tilde{D}\tilde{D}\tilde{D}\}, \tag{E.5}$$

$$W_{12} = \tilde{C} + \tfrac{1}{2}\{\tilde{C}\tilde{D}\} + \tfrac{1}{6}\{\tilde{C}\tilde{D}\tilde{D}\}, \qquad W_{13} = \{\tilde{C}\tilde{C}\} + \tfrac{1}{3}\{\tilde{C}\tilde{C}\tilde{D}\}, \tag{E.6}$$

$$W_{21} = \tilde{B} + \tfrac{1}{2}\{\tilde{B}\tilde{D}\} + \tfrac{1}{6}\{\tilde{B}\tilde{D}\tilde{D}\}, \qquad W_{31} = \tfrac{1}{2}\{\tilde{B}\tilde{B}\} + \tfrac{1}{6}\{\tilde{B}\tilde{B}\tilde{D}\}, \tag{E.7}$$

$$W_{22} = \tilde{A} + \tfrac{1}{2}(\{\tilde{B}\tilde{C}\} + \{\tilde{A}\tilde{D}\}) + \tfrac{1}{6}(\{\tilde{C}\tilde{B}\tilde{D}\} + \{\tilde{A}\tilde{D}\tilde{D}\}), \tag{E.8}$$

$$W_{23} = \{\tilde{A}\tilde{C}\} + \tfrac{1}{3}(\{\tilde{A}\tilde{C}\tilde{D}\} + \{\tilde{C}\tilde{C}\tilde{B}\}), \qquad W_{32} = \tfrac{1}{2}\{\tilde{A}\tilde{B}\} + \tfrac{1}{6}(\{\tilde{A}\tilde{B}\tilde{D}\} + \{\tilde{B}\tilde{B}\tilde{C}\}), \tag{E.9}$$

$$W_{33} = \{\tilde{A}\tilde{A}\} + \tfrac{1}{3}(\{\tilde{A}\tilde{B}\tilde{C}\} + \{\tilde{A}\tilde{A}\tilde{D}\}). \tag{E.10}$$

The block W_{11} contains the on-site evolution through third order in the Taylor expansion. The blocks W_{12} and W_{13} open the single- and doubled-memory sectors, while W_{21} and W_{31} close them. The remaining blocks propagate interactions across the bond and combine $\tilde{A}$ with the required symmetrised products. The doubled-memory sector permits two interactions to cross the same bond simultaneously and contains nine of the thirteen virtual states. It therefore represents the overlapping pairs generated by the NNN coupling that are absent from W^{II}.

E.2 Comparing the two constructions in practice

We first examine how well each construction preserves the norm under repeated application. We apply each operator to $|X+\rangle^{\otimes N}$ on a chain of $N = 100$ sites at $p = 0.1$, using $\delta t = 0.05$ and truncating the state at 10^{-12} after every layer:

	W^{I}	W^{II}	*VD2*
virtual bond	4	4	13
$\lVert\psi\rVert$ after 1 layer	1.3514	1.1341	0.99995
$\lVert\psi\rVert$ after 8 layers	12.6146	3.0181	0.99923
$\lVert\psi\rVert$ after 16 layers	307.3900	10.8328	0.99751

The norm error accumulates rapidly under the two first-order constructions. After sixteen layers, W^{I} increases the norm by more than two orders of magnitude and W^{II} by approximately one order of magnitude. By contrast, the *VD2* norm remains within 2.5×10^{-3} of unity. A calculation at $T = 10$ contains approximately one hundred temporal layers, so the first-order operators are unsuitable at this step size.

Norm drift compares the accumulated stability but does not determine the largest usable time step at a fixed accuracy. We therefore compare the state error at a fixed physical time. For $N = 12$, the Hamiltonian can be diagonalised exactly. We evolve $|X+\rangle^{\otimes N}$ to $T = 1$ using $T/\delta t$ layers and compare the result with the exact state at $p = 0.1$:

	W^{I}		W^{II}		*VD2*	
δt	error	slope	error	slope	error	slope
0.2	2.01×10^{1}	–	2.54	–	1.09×10^{-1}	–
0.1	4.43	2.18	9.29×10^{-1}	1.45	2.44×10^{-2}	2.16
0.05	1.36	1.71	3.93×10^{-1}	1.24	5.93×10^{-3}	2.04
0.025	5.31×10^{-1}	1.35	1.80×10^{-1}	1.12	1.47×10^{-3}	2.01
0.0125	2.36×10^{-1}	1.17	8.65×10^{-2}	1.06	3.68×10^{-4}	2.00
0.00625	1.12×10^{-1}	1.08	4.24×10^{-2}	1.03	9.24×10^{-5}	2.00

The slope is the effective convergence order obtained from each error and the result at the preceding step size. Because the error contains higher-order terms, this estimate approaches the asymptotic order as δt decreases. The *VD2* slope is 2.04 at $\delta t = 0.05$ and remains 2.00 at smaller steps. The corresponding slopes for W^{I} and W^{II} are 1.71 and 1.24 at $\delta t = 0.05$, approaching 1.08 and 1.03 only at the smallest step. These results confirm second-order convergence for *VD2* and first-order convergence for the other two constructions.

At $p = 0.5$, W^{II} approaches the asymptotic regime more slowly, with its measured slope decreasing from 1.85 to 1.07 over the sampled steps. The stronger interaction therefore requires a smaller δt before the first-order term dominates. The *VD2* slope remains close to 2.00 throughout. At the next halved step, its error is 2.7×10^{-5} and the measured slope decreases to 1.77 because state truncation and the finite accuracy of the exact reference become comparable to the evolution error.

The larger virtual bond of *VD2* is compensated by the larger time step allowed at fixed accuracy. A first-order construction requires a smaller δt, which lengthens the temporal chain in inverse proportion and introduces an additional truncation at every new site. At $T = 10$, the temporal chain contains 104 sites for $\delta t = 0.1$ and 204 sites for $\delta t = 0.05$. The production calculations therefore use *VD2* with $\delta t = 0.1$.

F Implementation details of the block power method

Section 6.2 introduces the block power method, in which k left and k right temporal states are propagated through the transfer matrix and the eigenproblem is projected onto their span. This appendix gives the implementation of that procedure, the numerical controls used in production, and the tests performed to assess its reliability. The routine `block_transfer_eigs` is built on `ITransverse.jl` [36]; the source code, analysis notebooks, and cluster drivers are available in the thesis repository [61].

F.1 The iteration as implemented

Each iteration updates the blocks $\{|R_j\rangle\}$ and $\{\langle L_j|\}$ through four operations: application, projection, solution of the reduced problem, and separation of the resulting states.

We first apply the transfer matrix to each right state and its transpose to each left state, requiring $2k$ operator–state products. Each application increases the temporal bond dimension, so the resulting state is immediately truncated to the bond cap used in that iteration. We then construct the overlap and projected-transfer matrices of Eq. (19),

$$S_{ij} = \langle L_i | R_j \rangle \,, \qquad M_{ij} = \langle L_i | \, \mathcal{E} \, | R_j \rangle \,. \tag{F.1}$$

Both use the same bilinear contraction without complex conjugation, since the left states are already represented as bras. Figure 20 shows the two contractions, and Algorithm 1 summarises one complete iteration.

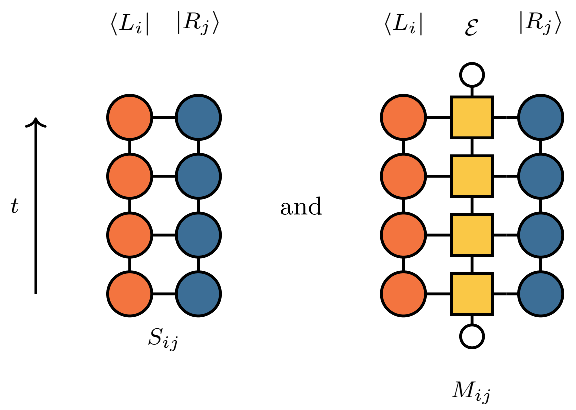


Figure 20: **The two projected matrices.** Contractions defining the $k \times k$ matrices of Eq. (19). Each tMPS forms a column, while $\mathcal{E}$ is the corresponding column of the space–time network closed at both ends by the product state (small open circles). The left chain enters directly as a bra, without complex conjugation.

Algorithm 1 Block power method for the k dominant eigenpairs of $\mathcal{E}$.

$|R_j\rangle\,,\ \langle L_j| \leftarrow$ random complex tMPSs, $\quad j = 1, \dots, k$
repeat
  $\left|\tilde{R}_j\right\rangle \leftarrow \mathrm{truncate}_\chi\left(\mathcal{E}\,|R_j\rangle\right), \qquad \left\langle\tilde{L}_j\right| \leftarrow \mathrm{truncate}_\chi\left(\langle L_j|\,\mathcal{E}\right)$ ▷ apply
  $S_{ij} \leftarrow \langle L_i|R_j\rangle, \qquad M_{ij} \leftarrow \left\langle L_i\middle|\tilde{R}_j\right\rangle = \langle L_i|\,\mathcal{E}\,|R_j\rangle$ ▷ project
  solve $Mv = \theta\, Sv$ for Ritz values $\theta_1, \dots, \theta_k$ and coefficients V, U
  $|R_j\rangle \leftarrow \mathrm{truncate}_\chi\left(\sum_i V_{ij}\left|\tilde{R}_i\right\rangle\right), \qquad \langle L_j| \leftarrow \mathrm{truncate}_\chi\left(\sum_i U_{ij}\left\langle\tilde{L}_i\right|\right)$ ▷ separate
  normalise every block member
until the leading Ritz values stop changing within tolerance

The reduced problem $Mv = \theta Sv$ becomes ill conditioned when the block states approach linear dependence. Direct inversion of S would then amplify its smallest singular values, which are dominated by numerical noise. We instead form the Moore–Penrose pseudoinverse S^+, retaining only singular values larger than 10^{-12} times the largest one. The discarded directions are mapped to zero, giving

$$W = S^+ M, \qquad W\, v_j = \theta_j\, v_j. \tag{F.2}$$

The eigenvalues θ_j are the Ritz approximations to $\mu_0, \dots, \mu_{k-1}$ and are ordered by decreasing modulus. Writing the right coefficient vectors as the columns of V and the Ritz values as the diagonal matrix Θ gives $W = V\Theta V^{-1}$.

The left coefficients must be paired with the same Ritz values as the corresponding right coefficients. Solving a separate transposed problem would require matching two independently ordered spectra, which becomes ambiguous inside a near-degenerate cluster. We instead derive the left coefficients from the decomposition of W. For $u^\mathsf{T} M = \theta u^\mathsf{T} S$, define $y^\mathsf{T} = u^\mathsf{T} S$. On the retained subspace, $S^+ S = \mathbb{1}$, and hence

$$u^\mathsf{T} M = y^\mathsf{T} S^+ M = y^\mathsf{T} W. \tag{F.3}$$

Thus y^T is a left eigenvector of W. Since the left eigenvectors are the rows of V^{-1}, the coefficients are

$$u_j = (S^+)^\mathsf{T} (V^{-1})^\mathsf{T} e_j, \tag{F.4}$$

where e_j is the j-th unit vector. The common index j fixes the left–right pairing directly. It also gives biorthogonality in the S metric, $u_i^\mathsf{T} S v_j = \delta_{ij}$.

The final operation reconstructs the states from these coefficients. Each right state is formed as $\sum_j (v_i)_j \, |AR_j\rangle$ from the propagated block, with the analogous construction on the left. We sum the tensors exactly and compress the result afterwards. Density-matrix addition of the MPSs is cheaper but becomes unstable when the states are nearly parallel, precisely where separation is most difficult. Each reconstructed state is renormalised. If its norm underflows or overflows, the state is replaced by a new random vector so that the lost direction is not propagated as numerical noise.

F.2 The eigenvalue-only mode

Separating the invariant subspace into individual eigenvectors is the least stable part of the iteration. As two eigenvalues approach, the coefficient matrix V becomes ill conditioned and amplifies the truncation noise in the reconstructed states. The Ritz values are less sensitive because they depend on the projected operator over the complete subspace rather than on a particular basis within it.

When only the spectrum is required, we therefore propagate orthonormal bases instead of separated eigenvectors. After ordering the Ritz values by modulus, the coefficient matrices are replaced by the orthonormal factors from their QR decompositions,

$$V \to Q_V, \qquad U \to Q_U, \qquad Q^\dagger Q = \mathbb{1}. \tag{F.5}$$

The first j columns of each Q span the same subspace as the corresponding first j coefficient vectors. The ordering of the leading subspaces is therefore retained without assigning an individual eigenvector to each block member.

This basis change leaves the Ritz values invariant. Under invertible recombinations

$$|R_j\rangle \to \sum_i B_{ij} \, |R_i\rangle \,, \qquad \langle L_j| \to \sum_i C_{ij} \, \langle L_i| \,, \tag{F.6}$$

the projected matrices transform as $S \to C^\mathsf{T} S B$ and $M \to C^\mathsf{T} M B$. The reduced problem becomes

$$C^\mathsf{T} M B \, \tilde{v} = \theta \, C^\mathsf{T} S B \, \tilde{v}, \qquad \tilde{v} = B^{-1} v, \tag{F.7}$$

and therefore has the same θ. The QR rotation changes only the basis within the subspace. Its benefit is numerical: Q_V and Q_U are unitary, so the states passed to the next truncation are not amplified by the ill-conditioned eigenvector transformation.

We compared both modes at $p = 0.1$ using the same initial random states and otherwise identical settings. Over $T = 2$–8, the relative difference in the leading modulus ranges from 6×10^{-7} to 2×10^{-5}, while the ratio of the first subleading modulus to the leading one agrees within 10^{-8}–10^{-4}. The less-converged members further down the block differ by 10^{-4}–10^{-2}. The two production arms at the same coupling, which also differ in other settings, show comparable ranges. The leading eigenvalue used to extract the central charge is therefore insensitive to the basis choice within the tested window, whereas the lower block members are not.

Both modes include safeguards against loss of rank. Every reconstructed state is renormalised, and a state with an underflowed or overflowed norm is replaced by a random vector. We also monitor the condition number of S. When it exceeds 10^{10}, the final block member is replaced by a random state made biorthogonal to the retained members using the bilinear form of Eq. (19). This restores an independent direction without modifying the converged part of the block.

Convergence is assessed from the two leading Ritz values. Before measuring their change, each value is matched to its nearest predecessor in the complex plane rather than compared at the same position in a list sorted by modulus. This matching is required because the leading moduli approach one another under emergent dual unitarity. At $p = 0$ and $T = 20$, the three largest moduli differ by only three parts in a thousand, while their phases remain distinct. Truncation noise can therefore exchange their order between iterations even when the individual Ritz values have stabilised. Position-by-position comparison would interpret such an exchange as a failure to converge. The iteration stops when the matched change falls below 10^{-6} or after 150 iterations without further improvement.

F.3 Validation against exact diagonalisation

The block method introduces two approximations: only k directions of the transfer matrix are retained, and every propagated state is truncated to finite bond dimension. Since neither approximation provides an accessible error bound, we validate the implementation against dense diagonalisation at short evolution times.

At short evolution times, the rotated transfer matrix is small enough to construct as a dense array and diagonalise exactly. This calculation involves neither state truncation nor subspace projection and gives the exact spectrum of the operator supplied to the block method.

Table 4 compares the two leading eigenvalues in three cases. The first uses the production construction at $p = 0.1$. The second uses a reduced three-site operator built from the first-order MPO of Section 5; its temporal sites have dimension three, and the block does not reach its bond cap, isolating the reduced solver from truncation effects. The third is the critical Ising chain, whose transfer matrix is independently known. The largest discrepancy in either modulus is 3.6×10^{-7} and occurs for the largest operator, consistent with truncation rather than the reduced eigensolver setting the accuracy.

This comparison validates the complete short-time pipeline: construction of the evolution operator, extraction of the rotated column, projection onto the block, and solution of the reduced problem. It does not test the late-time regime, where the leading eigenvalues become nearly degenerate, because the dense matrix grows exponentially with evolution time.

operator	dimension	exact $\lvert\mu_0\rvert$, $\lvert\mu_1\rvert$	error
ANNNI-type, $p = 0.1$	2197	0.9405, 0.9082	3.6×10^{-7}
ANNNI-type, $p = 0.1$, reduced test operator	27	0.9758, 0.3292	4.6×10^{-13}
critical Ising	256	1.5488, 0.0611	4.2×10^{-10}

Table 4: **Validation against exact diagonalisation.** Comparison between the block method and dense diagonalisation of the same rotated transfer operator at short evolution times. The reported error is the largest absolute deviation in the moduli of the two leading Ritz values.

F.4 Computational cost

The computational cost of one iteration has three contributions. Applying the transfer matrix requires $2k$ operator–state products, each of which multiplies the temporal bond dimension by the physical dimension of the column before truncation. Constructing S and M requires k^2 scalar contractions. The reduced $k \times k$ eigenproblem is negligible at the block sizes used here. Reconstructing the left and right blocks requires $2k$ linear combinations of k states. This last operation dominates the peak memory and runtime because an exact sum temporarily carries the combined bond dimensions of its terms before compression. The single-vector iteration of Section 6.1 forms no such combination, which is the main reason it is the cheaper route to the dominant pair, and why we use it wherever the eigenvectors themselves are needed.

We limit this temporary growth by truncating partial sums to the production bond dimension, leaving only the final addition exact. In the parameter range studied, the temporal bond dimensions remain below this cap, and calculations with and without the safeguard give the same leading eigenvalue and runtime. Its role is therefore protective rather than performance-critical.

A separate stopping rule limits iterations that no longer improve the Ritz values. With RTM truncation, the change in the leading eigenvalues can reach a noise floor above the formal tolerance. Without this rule, the iteration would continue after the values had stabilised. We therefore stop after 150 iterations without further improvement.

F.5 Numerical parameters and controls

The following settings affect the numerical path and attainable accuracy but not the target fixed point. Unless stated otherwise, the tests use the production parameters of Section 7.

Temporal bond dimension χ. Increasing the cap from 32 to 64 and 128 at a fixed random start leaves the iteration count, stopping condition, eigenvalues, and entropy unchanged. The temporal states remain well below the cap, so the singular-value cutoff rather than χ controls the truncation. Within reliable runs, the plateau—the constant value reached by the imaginary part of the temporal entropy in the middle of the strip—changes by a small fraction of the Ising central charge between $\chi = 64$ and 128, while the real-part slope changes by several times more. This difference supports using the plateau as the entropy estimator.

Singular-value cutoff. We compare cutoffs of 10^{-8}, 10^{-10}, and 10^{-12} across independent initial states. The eigenvalues agree to the precision reported in this thesis. The cutoff primarily changes the stopping condition: at the production value, the leading-eigenvalue change often reaches a floor above the tolerance, whereas tighter cutoffs more frequently satisfy it. The random initialisation also affects which outcome occurs. Within the tested range, the cutoff changes the convergence path but not the reported values.

Truncation scheme. The RTM and RDM schemes of Section 3.4 reproduce the same entropy plateau and leading eigenvalue wherever both were tested. The RDM scheme reaches the convergence tolerance at every tested time, whereas the RTM scheme can stop improving above it. The RDM calculations are approximately an order of magnitude more expensive, so we use RTM in production and retain RDM as a control.

Block size k. The required block size is determined by the number of states rather than the number of distinct scaling dimensions, since conformal descendants produce degeneracies within a tower. The fixed-boundary Ising tower begins $2, 3, 4, 4, 5, 5, \dots$ (Appendix G.1). A four-state block cuts through the degenerate pair at 4: independent ladders then agree on the leading eigenvalue but disagree on the first gap at approximately half of the evolution times. With $k = 8$, the retained block ends in a genuine spectral gap and this disagreement disappears. We therefore use $k = 8$ for the boundary-spectrum and production-tower calculations. The implementation can enlarge a block in stages if a near-degenerate cluster occupies the retained subspace, although this fallback was not required for the reported data.

Warm and cold starts. Each evolution time can be initialised from random states or from the converged block at the preceding time, re-indexed onto the longer temporal lattice. Both procedures give the same values at matched times. Warm starting did not reduce the iteration count in a test with unit time increments. Production ladders use half-unit increments, for which the preceding block provides a closer initial subspace.

Random initialisation and repetition. Independent random starts of the single-vector iteration can converge to different eigenvectors at long evolution times. Entropy calculations are therefore repeated until independent runs agree, as discussed in Section F.6. The leading eigenvalues are substantially less sensitive to this effect.

Selection of the physical branch. At each time we take the eigenvalue of largest modulus, using continuity with the preceding accepted point only to resolve differences below 1%, as in Section 6.3. Largest modulus is not always the right choice, because the transfer matrix is not normal: values computed from an incompletely converged subspace can lie outside the spectrum altogether, and a larger block contains more of them. Continuity is what detects this. If the selected modulus jumps away from the previous accepted value, we take that as a sign that the largest candidate is not physical, and reselect from the stored spectrum among the candidates lying within a few per cent of the previous value. In the eight-state boundary ladders of Appendix G.1 this happened at three times, and in each of them the physical eigenvalue was the next candidate down. Where an independent repetition exists for comparison, the reselected point agrees with it to within the reproducibility of the method.

Cooling regulator β_0. The imaginary-time intervals at the ends of the temporal chain add N_β sites and regularise the initial state. Changing β_0 rescales the leading eigenvalue through a non-universal normalisation, which cancels in phase differences. It also produces a substantial shift in the entropy plateau through $\epsilon_2 = 2\beta_0/T$. Dividing by the leading correction reduces the variation across the eight tested values of N_β by a factor of four to six, and the remaining difference decreases with T, as expected for an effect of order β_0/T. We therefore keep β_0 fixed within each ladder.

Trotter step δt. Halving δt while doubling N_β keeps β_0 fixed and leaves the entropy plateau unchanged within the reliable window. The point at which runs start to fail also remains at the same physical time. The accessible range is therefore not limited by the time discretisation at the tested steps.

Part of the eigensystem. Entropy calculations require separated left and right eigenvectors, whereas the spectral measurements require only an orthonormal basis for the leading invariant subspace. This difference, described in Section F.2, determines the distinct time ranges of the two routes in Section 7. The routine exposes this as a setting rather than as two separate codes: with it enabled, the iteration skips the separation into individual eigenvectors and the biorthogonal normalisation that only the entropy needs, and propagates orthonormal bases instead. Everything upstream is identical, which is what makes the comparison between the two modes a test of the basis alone.

F.6 The limits of the entropy route

The entropy and spectral routes require different information from the transfer matrix. The spectral analysis uses the Ritz values of the leading invariant subspace. The entropy requires the corresponding individual left and right eigenvectors. Eigenvalues obtained by subspace projection are second-order accurate in the error of the spanning vectors, whereas resolving individual eigenvectors becomes sensitive to the separation between neighbouring eigenvalues [62]. Emergent dual unitarity reduces these separations as the evolution time grows, so the entropy calculation loses reliability earlier.

We quantify this loss through independent random initialisations of the single-vector iteration, which produces every ensemble reported in this appendix. At each accessible time, we perform twenty runs for $p = 0$ and ten for the interacting couplings, using the production settings of Section F.5. A run is classified as failed when its imaginary plateau lies outside a physically admissible band around $\pi c/16$. Table 5 reports both a loose and a tight band to show the sensitivity to this criterion. Inside the quoted windows the two classifications differ only near the margins. They part company beyond them, where the surviving runs cluster at roughly half the expected plateau and the loose band accepts what the tight band rejects, so agreement between the two is itself a signal that a rung is still usable.

Failures occur at the level of individual runs. At a given time, failed runs return unphysical plateaux while the surviving runs remain consistent with neighbouring times. The failure probability increases with T, and the first failures move from $T = 18$ at $p = 0$ to $T = 4$ at $p = 0.5$. At $p = 0.3$ and $T \geq 9$, even the surviving plateaux span a factor of approximately 2.5. Once failures begin, an ensemble is therefore needed both to reject failed runs and to estimate the spread of the remaining values.

p	T	failed (loose)	failed (tight)	plateau of surviving runs
0	8–17	0/20	0/20	0.1252–0.1137
0	18	1/20	1/20	0.1144
0	20	0/20	1/20	0.1146
0	22	2/20	2/20	0.1080
0	24	4/20	6/20	0.1205
0.1	11	2/10	2/10	0.1199
0.1	12	0/10	2/10	0.1172
0.1	13	0/10	0/10	0.1158
0.3	4–6	0/10	0/10	0.1333–0.1264
0.3	7	2/10	2/10	0.1270
0.3	8	4/10	5/10	0.1184
0.3	9	6/10	7/10	0.1054
0.3	10	6/10	7/10	0.1007
0.5	3	0/10	0/10	0.1354
0.5	4	3/10	3/10	0.1351
0.5	5	7/10	7/10	0.1310
0.5	6	10/10	10/10	–

Table 5: **Failure rate of the single-vector iteration.** Results from independent random initialisations at each evolution time. A run is classified as failed when its imaginary plateau lies outside a band around $\pi c/16 = 0.098$. The loose band is $(0.05, 0.20)$ and the tight band is $(0.07, 0.15)$. Consecutive times without failures are grouped; the final column gives the median plateau of the surviving runs.

The stopping condition does not distinguish these failures. With RTM truncation, the change in the leading eigenvalues decreases initially but then levels off above the convergence tolerance at nearly every time beyond $T \approx 10$, for both correct and failed runs. The truncation continually introduces errors set by its cutoff, so reaching this floor does not show that the physical result is unreliable. Figure 21 shows this at a single point. The change in the leading Ritz values falls by five orders of magnitude, then stops falling and stays above the tolerance until the no-improvement rule ends the run. Repeating the same point with RDM truncation, from the same random states, the same quantity continues past the tolerance and the iteration terminates normally. The two runs return the same leading eigenvalue to six decimal places, so the floor belongs to the truncation scheme and not to the transfer matrix, and a run that ends on the no-improvement rule has not necessarily failed to find the answer.

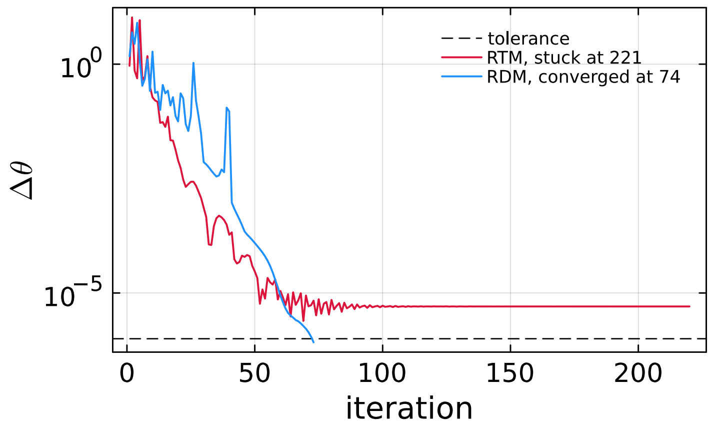


Figure 21: **How the iteration stops.** The change $\Delta\theta$ in the two leading Ritz values between successive iterations, at $p = 0.1$ and $T = 4$, for the two truncation schemes started from the same random states. Under RTM truncation the change levels off above the convergence tolerance and the run ends after 150 iterations without improvement; under RDM truncation it crosses the tolerance and the run stops there. The two return the same leading eigenvalue to six decimal places.

Table 6 compares a subset of points with RDM truncation. The RDM iteration reaches the same tolerance at every tested time, including $T = 24$ at the integrable point, and reproduces the RTM plateau within the observed spread. The noise floor is therefore associated with the RTM truncation rather than the transfer matrix. The RDM calculations are approximately an order of magnitude more expensive, which motivates using RTM for production and RDM as a control.

The eigenvalue-only block iteration shows no comparable seed dependence in the tests performed at $p = 0.3$. Six independent starts at $T = 6$ and 6.5, the end of the fitting window in Table 1, give leading moduli agreeing within 3×10^{-5} and 10^{-4}, respectively. Their phases agree within 8×10^{-5} and 1.3×10^{-4}, and both points agree with the corresponding cluster calculation within 10^{-4}. At $T = 7$, where two of ten entropy runs fail, all six block runs again return consistent leading eigenvalues, with phase differences below 3×10^{-4}. Within this test, the eigenvalue-only mode remains reproducible after the entropy route begins to fail.

The strong coupling dependence of the eigenvector conditioning is already present without tensor-network truncation. At short times, we construct the rotated transfer operator as a dense matrix and diagonalise it exactly, following Section F.3. Table 7 compares the condition number of the full eigenvector matrix at matched dense dimension, without

p	T	plateau		stopping		time (s)	
		RTM	RDM	RTM	RDM	RTM	RDM
0	8	0.1252	0.1248	converged	converged	12	147
0	12	0.1191	0.1211	no improvement	converged	53	785
0	14	0.1186	0.1185	no improvement	converged	98	1466
0	17	0.1137	0.1162	no improvement	converged	177	2627
0	20	0.1146	0.1145	no improvement	converged	283	5177
0	22	0.1080	0.1134	no improvement	converged	402	5125
0	24	0.1205	0.1126	no improvement	converged	460	7300
0.1	11	0.1199	0.1207	no improvement	converged	1316	43692

Table 6: **Comparison of the two truncation schemes.** The RTM entries are medians over the surviving seed ensembles in Table 5. The RDM entries are single runs, except at $T = 17$, where three seeds give 0.1153–0.1165. The stopping columns record whether the run reached the 10^{-6} tolerance or stopped after 150 iterations without improvement. Times are median runtimes per run at identical settings.

cooling sites and with $\delta t = 0.5$. From $p = 0$ to the interacting couplings, the condition number increases by seven to nine orders of magnitude, while the distance between the two leading eigenvalues remains comparable. The poor conditioning at $p \neq 0$ therefore cannot be attributed solely to an approaching leading degeneracy. Its microscopic dependence on the interaction remains unresolved.

p	T	dimension	$\lvert\mu_0 - \mu_1\rvert$	$\mathrm{cond}(V)$
0	3.5	2187	0.21	3.0×10^2
0.1	1.5	2197	0.30	2.2×10^{11}
0.3	1.5	2197	0.19	2.1×10^{10}
0.5	1.5	2197	0.14	9.6×10^9

Table 7: **Conditioning of the exact eigenvector basis.** Exact diagonalisation of the rotated transfer operator at matched dense dimension, without cooling sites and with $\delta t = 0.5$. The fourth column gives the distance between the two leading eigenvalues, while $\mathrm{cond}(V)$ is the condition number of the complete eigenvector matrix. The strong coupling dependence is not explained by the leading spectral gap.

Three limitations remain. First, the mechanism by which the coupling controls the eigenvector conditioning is unknown. Second, no diagnostic from a single run reliably distinguishes a degraded surviving entropy result near the point at which the method starts to fail; independent repetitions remain necessary. Third, the block-method seed ensembles cover only one coupling at three evolution times. They establish reproducibility there but do not determine how the failure probability varies across the full parameter range. The spectral windows in Table 1 therefore continue to use the phase-increment criterion of Section 6.3.

G Supporting measurements and fitting conventions

This section collects the measurements and fitting conventions that support Section 7: the spectra for different boundary conditions, entropy profiles at the remaining couplings, the central-charge analysis from the real part of the entropy, and the fitting windows used for the quoted universal quantities.

G.1 The boundary spectrum and the choice of boundary conditions

The boundary dimensions in Section 7 are extracted for the initial state $|X+\rangle$. More generally, the states at the two ends of the temporal chain define conformal boundary conditions, and their pair determines the operator content allowed on the strip. We test this correspondence by changing the boundary states and verifying the predicted change in the transfer-matrix spectrum, following the analysis of the Ising chain by Bou-Comas et al. [23].

Which operators a pair admits follows from the geometry of the strip. When both ends carry the same condition, the strip may contain nothing at all, so the identity is allowed and the spectrum begins at $x_0 = 0$. When the two ends carry different conditions, something must change the boundary condition along the edge, and in a CFT that change is an operator in its own right, sitting where the two conditions meet; the identity cannot do it, since it would leave the boundary condition unchanged. The spectrum then begins at the dimension of the lightest operator that converts one condition into the other. The three Ising operators supply these values: the identity at 0, the energy at $\frac{1}{2}$ and the spin at $\frac{1}{16}$. Each brings its own descendants, spaced by integers, so the spectrum is a set of towers rather than a list of isolated values [27, 35].

The three Ising boundary conditions are represented by the product states $|X+\rangle$, $|Up\rangle$, and $|Dn\rangle$. The state $|X+\rangle$ realises the free condition because $\langle\sigma^z\rangle = 0$, while $|Up\rangle$ and $|Dn\rangle$ realise the two fixed conditions. Choosing the states independently at the two temporal boundaries produces four distinct pairs. When the two states differ, the network evaluates the transition amplitude $\langle b_2|\, U(T)\, |b_1\rangle$ rather than a return amplitude, but the contraction is otherwise unchanged.

Phase gaps determine the tower spacings $x_i - x_0$. Table 8 reports the leading spacings at the longest time reached for each boundary pair, and all four agree with their predicted towers. The two mixed-boundary cases have the same integer spacings because each tower is generated from a single primary operator.

The spacings alone do not determine the starting dimension x_0. The common non-universal factor $e^{-f\beta - f_s}$ in Eq. (A.13) cancels from μ_i/μ_0, together with x_0. The tower position is instead encoded in the $1/T$ coefficient of the leading phase. The bulk term $f\beta$ contributes linearly in T, while the real boundary term f_s changes only the modulus (Appendix A.2). The remaining coefficient is

$$\frac{\pi}{v}\left(x_0 - \frac{c}{24}\right). \tag{G.1}$$

At the integrable point, $c = \frac{1}{2}$ and $v = 2$ are known, so fitting Eq. (6) determines x_0. The two towers predicted to begin at zero give values within 0.002 of zero, while the mixed pairs are separated into the expected $\frac{1}{2}$ and $\frac{1}{16}$ sectors. The fitted phase coefficient is systematically 5–10% smaller in magnitude than its asymptotic prediction. The same deficit produces the spectral estimate $c \approx 0.45$ rather than 0.50 in Table 1, since both quantities are obtained from this coefficient. Two controls identify the leading sources. Adding the next order in $1/T$ raises the coefficient from 90% to 97% of the prediction and removes most of its dependence on the lower fit boundary. Varying the regulator at fixed fitting window gives a coefficient that decreases approximately linearly with β_0, from 95% of the prediction at the smallest regulator to 49% at the largest, and extrapolates to the expected value as $\beta_0 \to 0$.

The production regulator $\beta_0 = 0.2$ therefore introduces an approximately 10% downward bias in this coefficient. Figure 22 shows that dependence directly: the coefficient falls linearly with the amount of cooling and extrapolates to the conformal value as $\beta_0 \to 0$.

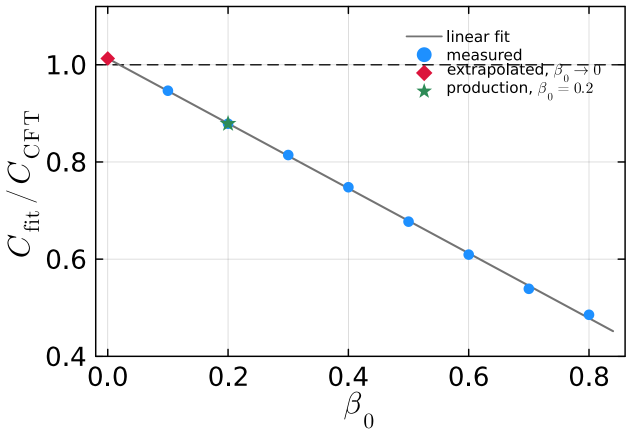


Figure 22: **The fitted phase coefficient against the cooling regulator**, at the integrable point, as a fraction of the value predicted by Eq. (6). Each point is a separate fit over the same window, differing only in N_β. The dependence is linear and extrapolates to within a few per cent of the prediction at $\beta_0 \to 0$, while the production choice $\beta_0 = 0.2$ sits about 10% below it. The deficit is therefore a finite-regulator effect and not a failure of the relation.

Both finite-time and regulator corrections act in the same direction, explaining why all fitted tower positions lie below their targets. We retain the same two-term fitting convention at every coupling because the shorter interacting windows cannot constrain an additional parameter. The remaining deficit is treated as a systematic bias rather than corrected point by point.

The eight-state calculations also resolve degeneracies among conformal descendants. For fixed boundaries, the predicted sequence is $2, 3, 4, 4, 5, 5$, and the numerical spectrum contains pairs at 4 and 5 within the accuracy of the individual estimates. The free–fixed spectrum shows the corresponding doubling one level earlier. These higher levels provide evidence for the tower structure but are not used as precision estimates of individual dimensions.

boundary pair	tower $x_i - x_0$	measured	x_0	measured
free–free	$\frac{1}{2},\ \frac{3}{2},\ 2$	$0.503,\ 1.488,\ 2.016$	0	0.002
fixed–fixed	$2,\ 3,\ 4$	$1.974,\ 2.982,\ 3.961$	0	0.002
fixed–antifixed	$1,\ 2,\ 3$	$0.994,\ 2.001,\ 3.045$	$\frac{1}{2}$	0.474
free–fixed	$1,\ 2,\ 3$	$0.988,\ 1.969,\ 2.980$	$\frac{1}{16}$	0.060

Table 8: **Boundary spectra for the four pairs of boundary conditions.** Results at the integrable point. The spacings are obtained from the phase gaps at the longest time reached by each ladder, while the tower position follows from fitting Eq. (6) to the leading phase over the complete ladder. The mixed pairs have identical spacings and are distinguished by x_0.

G.2 The entropy profiles at stronger coupling

Figure 24 extends the comparison of Section 7.1 to the remaining couplings. The real part follows the chord profile with $c = \frac{1}{2}$ and the imaginary part forms an interior plateau above $\pi c/16$. The number of accessible evolution times decreases with p, leaving the $p = 0.5$ ladder too short to constrain the asymptotic extrapolation. Near the two ends of the strip

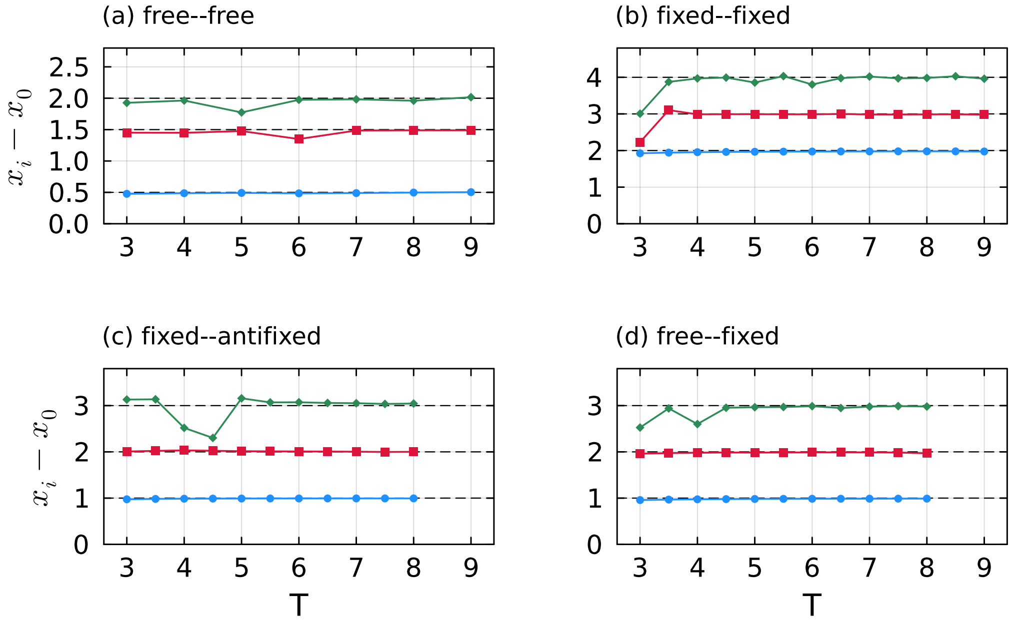


Figure 23: **Boundary towers at the integrable point.** The three leading spacings $x_i - x_0$ are obtained from Eq. (7); dashed lines mark the predicted values. (a) Free–free. (b) Fixed–fixed. (c) Fixed–antifixed. (d) Free–fixed. The two mixed cases have the same spacings and are distinguished by the tower positions in Table 8.

the imaginary part also rises above the interior plateau before falling away. Equation (8) applies to a cut with both endpoints in the bulk, and that condition fails as one endpoint approaches the boundary. The deviation comes mainly from the cooling region: since β_0 is held fixed while T grows, it occupies a fixed interval at each end rather than a fixed fraction of the strip, so these features stay confined to the edges, take up a smaller part of the profile at longer times, and lie outside the interior window from which we read the plateau.

G.3 The central charge from the real part

Both parts of the generalised entropy contain the central charge, but they do not provide equally stable estimators. The imaginary part approaches the single plateau value $\pi c/16$. The real part has slope $c/8$ in the chord variable and also contains the non-universal offset s_0.

The main limitation is the leading finite-time correction. For the Rényi-2 entropy in the Ising class, it decays as $w^{-1/2}$ in the chord variable [23]. Omitting this term gives apparent central charges of approximately 0.65–0.70, including at the integrable point where $c = \frac{1}{2}$ is known. We therefore fit

$$\operatorname{Re} S_2(t) = s_0 + \frac{c}{8} W(t,T) + a\, w(t,T)^{-1/2}. \tag{G.2}$$

Although this form describes the profiles, it does not determine c reliably. Over the accessible part of the strip, the logarithmic term and the correction have similar shapes. A fit with both c and a free is therefore poorly conditioned and cannot separate their contributions. Figure 25(a) illustrates this degeneracy: both the corrected form with $c = \frac{1}{2}$

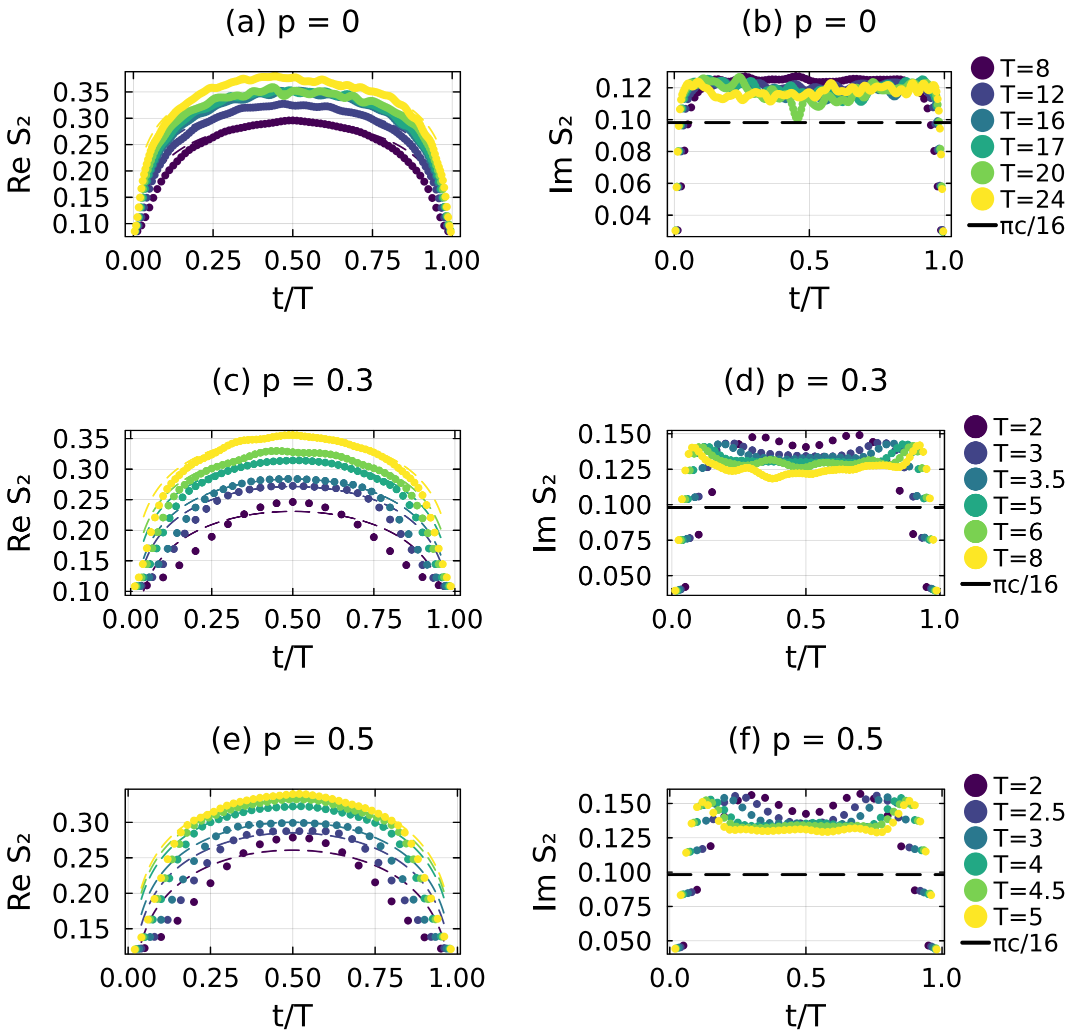


Figure 24: **Generalised temporal entropy at the remaining couplings.** Results for $p = 0$, 0.3, and 0.5, arranged by row. The left panels show $\mathrm{Re}\, S_2$ against t/T, with dashed curves from Eq. (8) at $c = \frac{1}{2}$ and a fitted non-universal constant. The right panels show $\mathrm{Im}\, S_2$, with the dashed line at $\pi c/16 = 0.098$. The offset from this line is the finite-time correction.

and an uncorrected fit with $c \approx 0.6$ describe the same profile. With c free, the estimate drifts with evolution time and crosses $\frac{1}{2}$ without approaching a plateau [Figure 25(b)].

The real part can instead test consistency with the expected value. For $T \gtrsim 12$, fixing $c = \frac{1}{2}$ and fitting only s_0 and a describes the profiles as well as the unrestricted fit. The fitted correction amplitude also agrees with that required by the imaginary part. The real and imaginary components are therefore consistent with the same finite-time correction.

We quote the imaginary plateau because it avoids the ill-conditioned separation of three contributions. After extrapolating its finite-time correction, the remaining value is $\pi c/16$. The real part must distinguish the logarithmic term, a similar-shaped correction, and the non-universal offset, which the accessible data do not constrain independently.

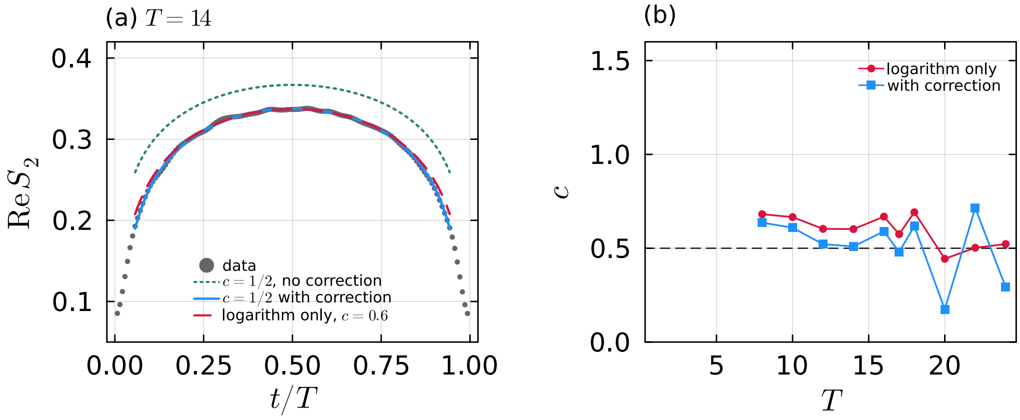


Figure 25: **Central charge from the real part of the entropy.** (a) An integrable-point profile fitted by the leading logarithm at $c = \frac{1}{2}$ (dotted), the corrected form of Eq. (G.2) at $c = \frac{1}{2}$ (solid), and the uncorrected form with c free (dashed). (b) Values of c obtained with and without the correction as a function of evolution time. Neither series converges within the accessible range.

G.4 Fit windows and conventions

The universal quantities in Table 1 are obtained from least-squares fits. This subsection specifies the fitting windows, fixed parameters, and finite-time correction used in those fits. Numerical controls on the underlying calculation are given in Section F.5.

The upper fitting boundary is determined from the physical branch. The phase increment of μ_0 is approximately constant while the branch remains resolved; a persistent departure from that behaviour sets the end of the window. The branch itself is tracked as described in Section 6.3. The lower boundary is a convention. Removing early times reduces the error of the $1/T$ expansion in Eq. (6) but also reduces the number of data points and raises the fitted central charge. We use the widest window permitted by the upper-bound criterion at every coupling. This common convention retains the most data and makes the dependence on the lower boundary explicit rather than selecting it separately for each coupling.

The ladders entering Eq. (6) are extended beyond the times covered by the block iteration using the leading eigenvalue of the single-vector method, whose modulus agrees with the block result to better than 10^{-5} wherever both are available. At those times the branch is selected by continuity of the phase advance rather than by modulus, and a rung is retained only if the root-mean-square residual of the fit does not increase when it is added.

The branch constant in Eq. (6) is fixed to $B = -\pi$, as required by $\lambda_0 = \log(-\mu_0)$. At the integrable point, a fit that includes the next order in $1/T$ and leaves B free gives $B/\pi = -1.0000$, with no detectable additional phase constant at the level of 10^{-4}. In a shorter two-term fit, allowing B to vary absorbs curvature associated with the central-charge term. This effect becomes severe over short windows, where the constant, $1/T$, and $1/T^2$ contributions cannot be separated. Fixing B removes this unconstrained degree of freedom. The phase is unwrapped over the full time series before the fitting window is selected so that the absolute branch is retained.

The sound velocity enters the inferred central charge and dimensions linearly. It is determined independently in Section 4.2; at the integrable point, the extrapolated value agrees with the exact result to five decimal places. Its uncertainty is therefore smaller than the other fit uncertainties considered here.

For the entropy plateau, we use the predicted Rényi-2 correction proportional to $T^{-1/2}$ and compare it with three alternatives: a constant, a $1/T$ decay, and a two-term form. Over the long integrable ladder, the constant gives the largest residual and the extrapolated value furthest from $\frac{1}{2}$. The predicted form and the two-term refinement give consistent asymptotes, showing that the resolved decay is not imposed solely by the chosen ansatz. Over short windows, all four forms describe the data similarly but extrapolate to widely different limits. Such windows do not constrain the asymptotic plateau.

The accessible range therefore determines how well an entropy extrapolation is constrained. We measure this at the integrable point by truncating its ladder to successively shorter windows. Ending at $T \leq 13$, 14, 18 and 24 returns 0.465, 0.495, 0.470 and 0.498, so a short window moves the answer by up to 0.04 and does so without a clear trend with the window length. The interacting ladders are shorter still, which is why the entropy-route values of Table 1 at $p = 0.3$ and $p = 0.5$ are quoted with their windows and read as indicative. The spectral route remains the more reliable of the two across the reported coupling range.